

Unblinding Request: 2021 Full Luminosity

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University



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LABORATORY

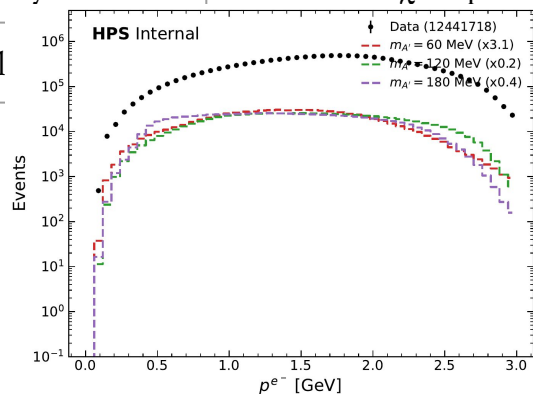
Unblinding 2021 Checklist

- I. Finalize Event Selection
- II. Background Model Validation
 - A. Quality Fits/Signal Extraction
 - B. Background Profiling
 - C. Viable Search Window for 2021
- III. Signal Model Choices
 - A. Window Range, signal template
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 - A. Asymptotic Likelihood
- V. No siggie? No Biggie! – upper limits
 - A. CLs to determine signal yield in individual datasets
 - B. CLs to determine shared coupling when combining datasets

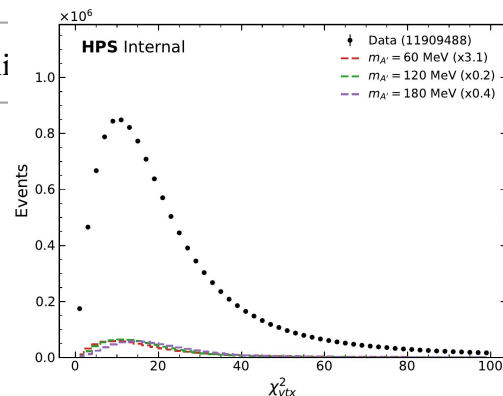
Results may be independent of coupling strength → We are looking at peaking structure over a QED background. This can be input to a variety of frameworks. Asymptotic probabilities and independent experiment hypotheses should be published and considered meaningful.

Event Selection (2021) (Pre-Selection+)

Selection	2021 preselection + prompt sample
Trigger / topology	Single-2 or Single-3; one selected good vertex
Momenta [GeV]	$p(\text{track}) > 0.4$; $p(e^-) < 2.9$; $p(\text{sum}) > 2.8$; $p(\text{vertex}) < 4.0$
Track quality / hits	$\chi^2(\text{track})/\text{ndf} < 20$; electron: 8 hits (updated selection)
Vertex quality	No vertex-fit χ^2 requirement
Positron cl	



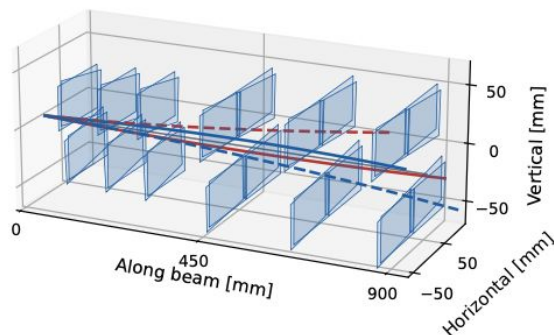
herited track-cluster tim



v5.0.5 preselection examples; these plots were not reprocessed with the updated eight-hit cut.

Acceptances by Geometry

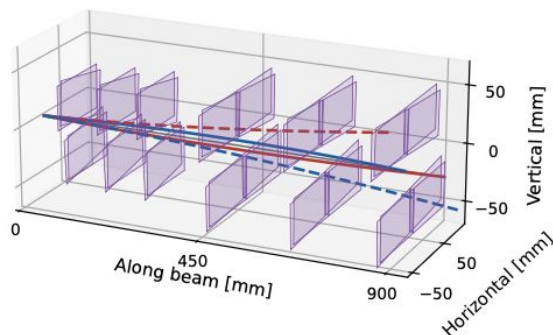
HPS 2015 | 1.056 GeV



Passing examples: 20 and 195 MeV

Nominal 15 mrad scale ≈ 15.84 MeV

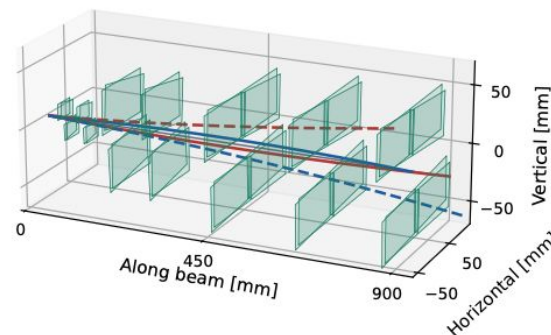
HPS 2016 | 2.300 GeV



Passing examples: 40 and 425 MeV

Nominal 15 mrad scale ≈ 34.50 MeV

HPS 2021 | 3.740 GeV



Passing examples: 60 and 715 MeV

Nominal 15 mrad scale ≈ 56.10 MeV

— Positron — Electron — Low-mass example - - - High-mass example

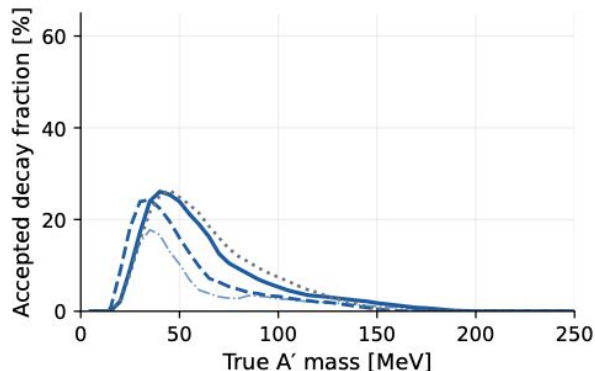
Note goes into acceptances more thoroughly, will start search in 2021 100% at 56 MeV

Acceptances by Campaign

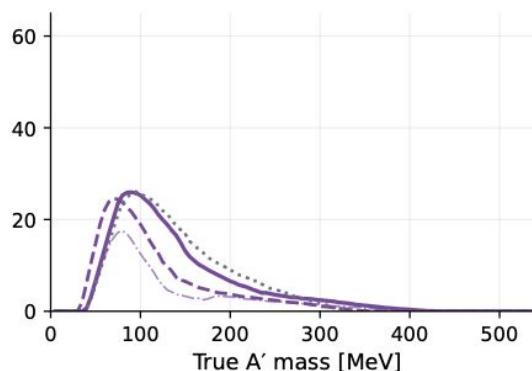
Run	$B_y(\mathbf{r}_{\text{map},0})$ [T]	Sampled masses [MeV]	Peak [MeV]	Peak fraction
2015	-0.2402	20–195	40	26.1%
2016	-0.5234	40–425	90	25.9%
2021	-0.8596	60–715	195	40.1%

— Field + hit cuts, $x = 1$ B = 0 + hit cuts, $x = 1$
--- Field + hit cuts, $x = 0.8$ -.-.- Field + all stations, $x = 1$

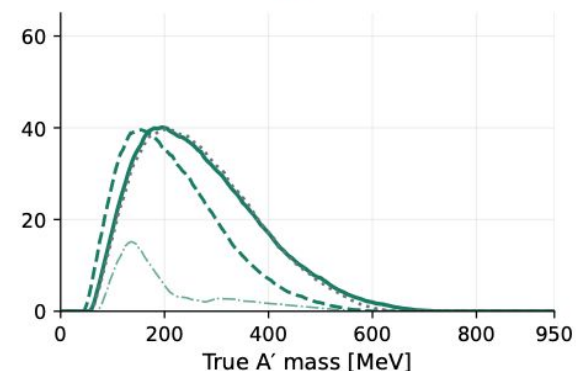
HPS 2015 | 1.056 GeV



HPS 2016 | 2.300 GeV



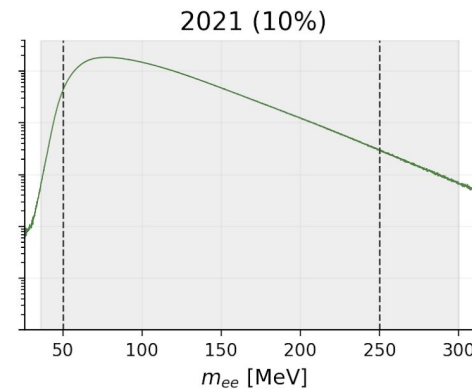
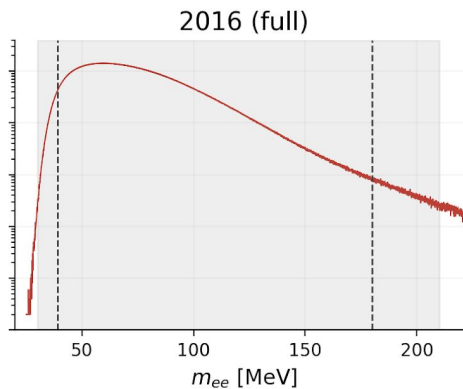
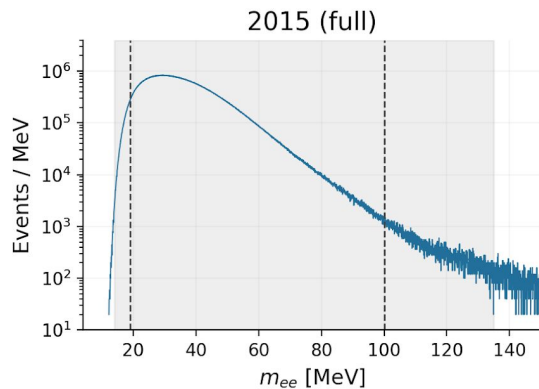
HPS 2021 | 3.740 GeV



HPS datasets for this phase

Run	Beam [GeV]	Sample	Search [MeV]	GP support [MeV]
2015	1.056	1170 nb ⁻¹	19–100	14–135
2016	2.3	10608 nb ⁻¹	39–180	30–210
2021	3.74	Native 10%	50–250	36–300

2021 native 10% is drawn from an approximately 160 pb⁻¹ parent run.



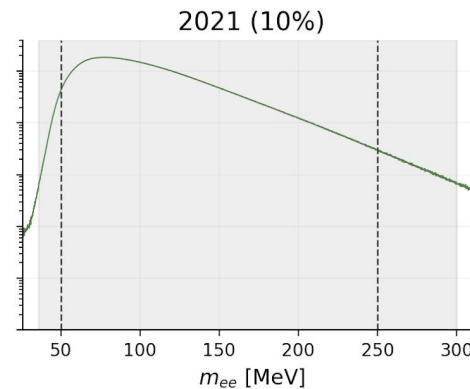
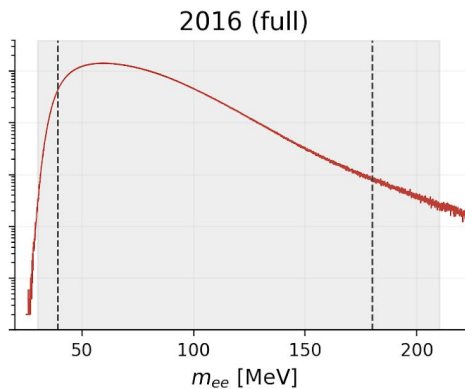
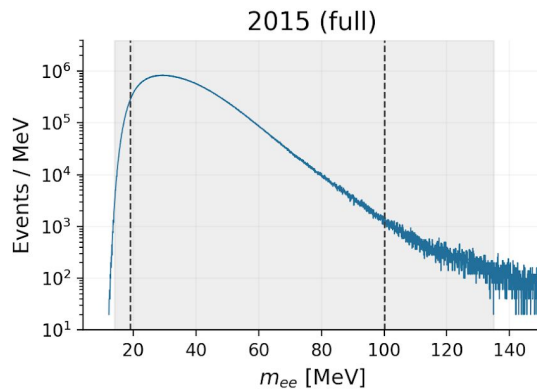
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2021 GP support begins at 36 MeV

2015 search extends to 100 MeV

Fit support supplies sidebands; the search range sets tested masses.



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Background Modeling: GPR

$$k(x, x') = C \exp \left[-\frac{(x - x')^2}{2\ell^2} \right]$$

ℓ : correlation distance in log mass

C : covariance amplitude

α : diagonal bin-noise variance $\approx 1/y$

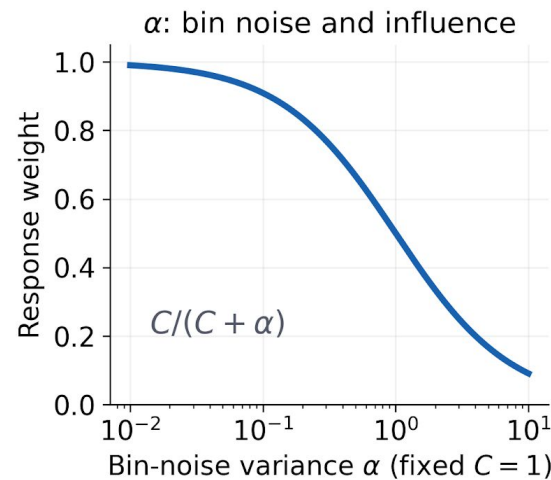
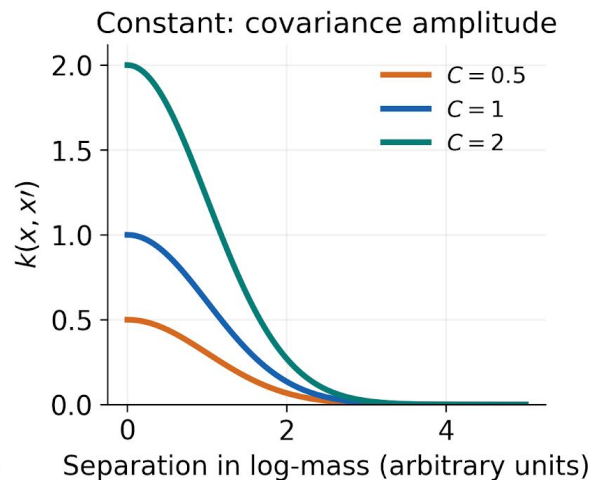
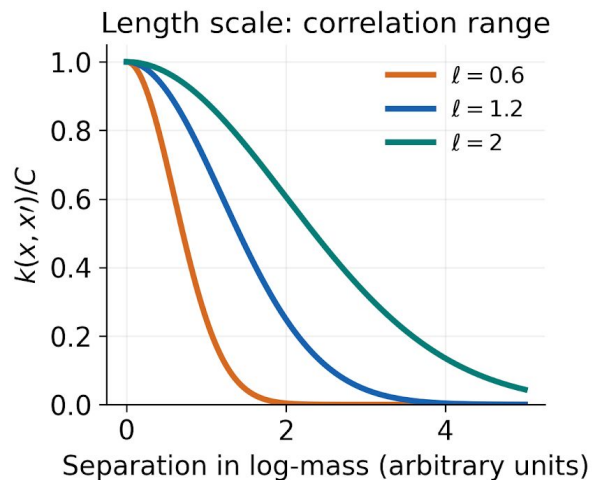


Illustration of separate GP roles; not a fit to HPS data. In the analysis: $x = \log m$ and $\alpha \approx 1/y$.

Background Profiling



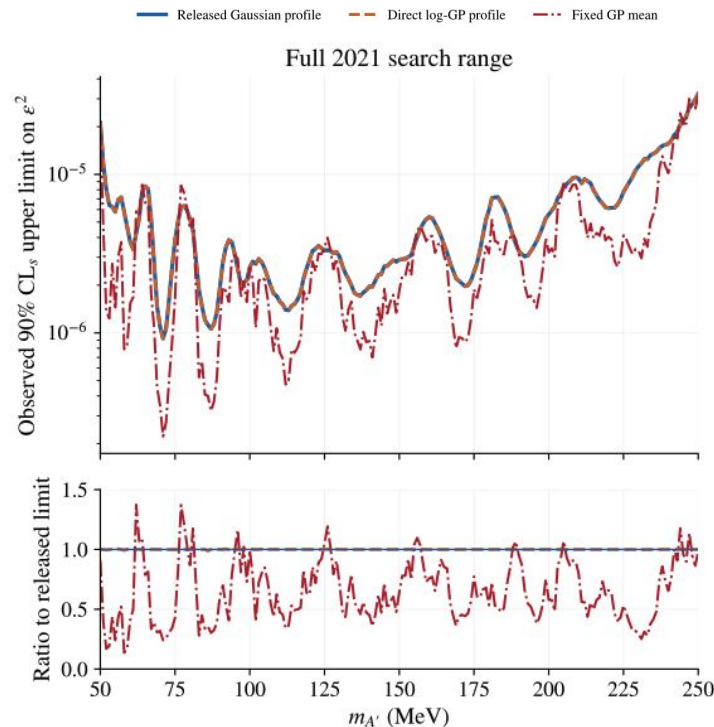
local Poisson mean

$$\lambda_d(A_d, \theta_d) = \underbrace{b_d}_{\text{projected bkg}} + \underbrace{L_d \theta_d}_{\text{profiled bkg}} + \underbrace{A_d w_d}_{\text{signal yield * template}}$$

profiled local likelihood

$$\mathcal{L}_d(A_d, \theta_d) = \left[\prod_{i=1}^{N_d(m)} \text{Pois}(n_{d,i} | b_{d,i} + (L_d \theta_d)_i + A_d w_{d,i}(m)) \right] \exp\left(-\frac{1}{2} \theta_d^\top \theta_d\right)$$

2021 10%: background profiling comparison



Background Profiling

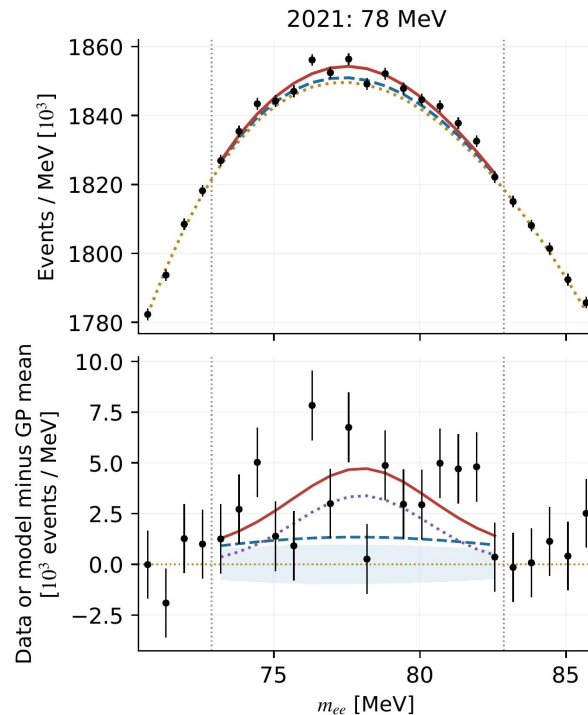


local Poisson mean

$$\lambda_d(A_d, \theta_d) = \underbrace{b_d}_{\text{projected bkg}} + \underbrace{L_d \theta_d}_{\text{profiled bkg}} + \underbrace{A_d w_d}_{\text{signal yield * template}}$$

profiled local likelihood

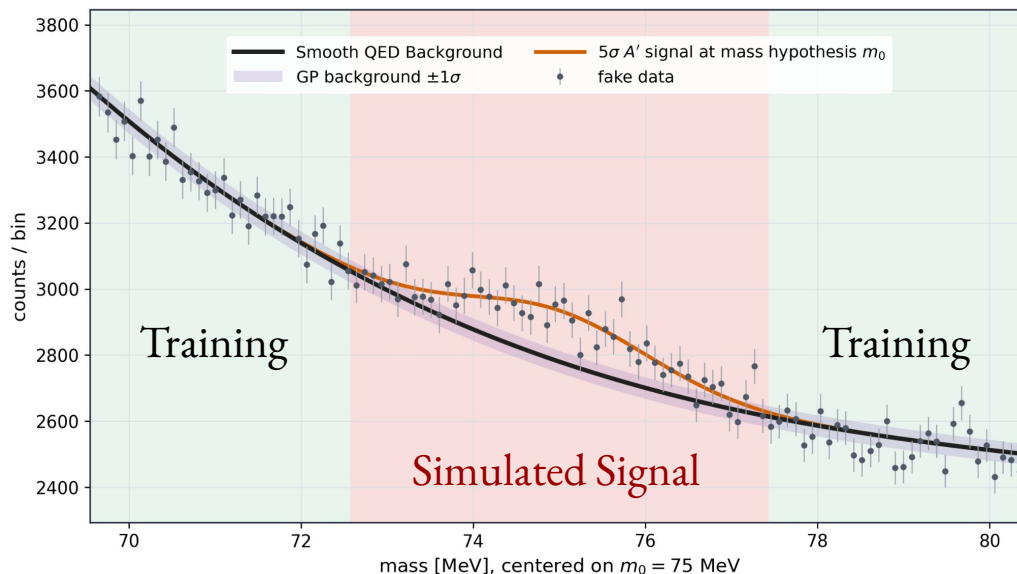
$$\mathcal{L}_d(A_d, \theta_d) = \left[\prod_{i=1}^{N_d(m)} \text{Pois}(n_{d,i} | b_{d,i} + (L_d \theta_d)_i + A_d w_{d,i}(m)) \right] \exp\left(-\frac{1}{2} \theta_d^T \theta_d\right)$$



Gold: GP mean | Blue: profiled background
Red: background + signal | Purple: signal

Prompt A' Signal Model and Backgrounds

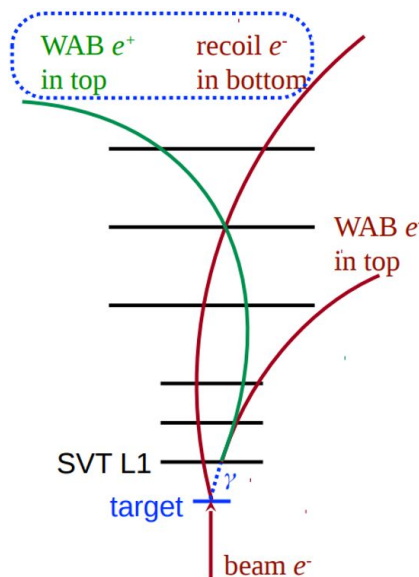
If A' exists within the acceptance of HPS, it will present itself as a **gaussian excess above background** in the IMD.



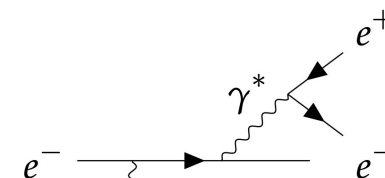
natural width of A' $\ll$ detector resolution
observed signal width = experimental mass resolution

Primary Backgrounds

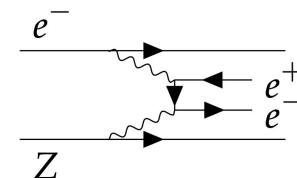
Wide Angle Brem



Radiative tridents

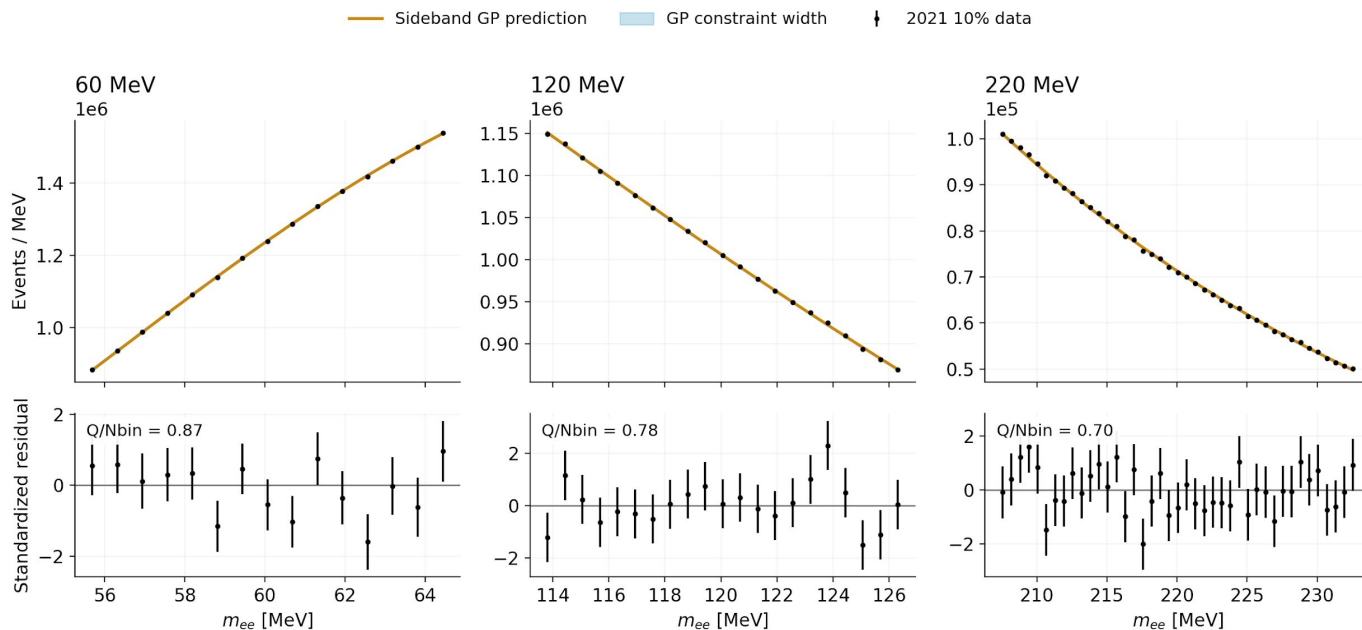


Bethe-Heitler tridents



Fitting the invariant mass distribution pt1

2021 native 10%: fixed GP predictions in three held-out windows

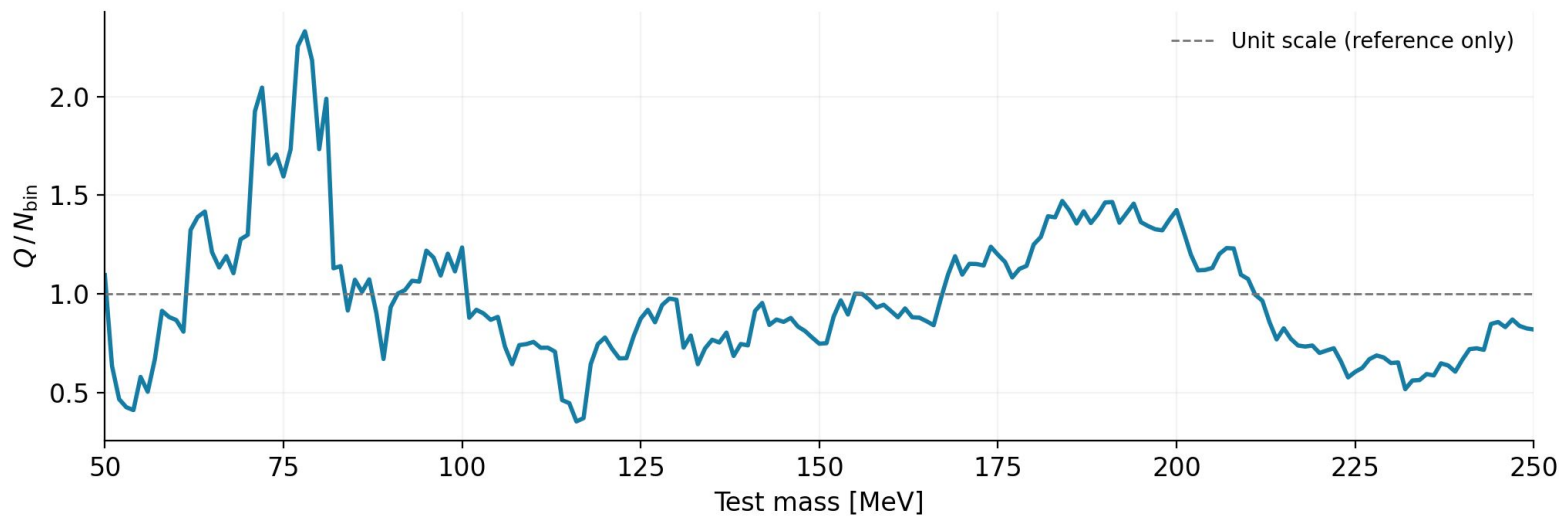


Archived kernel states; $\pm 2.25\sigma$ windows. Q/Nbin uses the full prediction covariance.

Fitting the invariant mass distribution pt2

Conditional residual size across the 2021 mass scan

$$Q = (\mathbf{n} - \mathbf{b})^T [\text{diag}(\mathbf{b}) + C_{GP}]^{-1} (\mathbf{n} - \mathbf{b})$$



Nbin counts held-out bins, not fitted degrees of freedom. Correlated windows; no calibrated χ^2 or KS p-value.

2021 functional form **fShiftSigPowTail** (excluding 65 MeV region)

$$B_d(m) = N_d S(m; t_d, w_d) (m - m_{0,d})^{a_d} \exp \left[-\frac{m - m_{0,d}}{\theta_d} + c_{1,d} u_d + c_{2,d} u_d^2 \right]$$

$$S(m; t, w) = [1 + \exp(-(m - t)/w)]^{-1} \quad u_d = (m - m_{\text{mid},d})/\tilde{m}_{\text{scale},d}$$

2015/2016 functional form **fGenGammaThresh**

$$G_d(m) = N_d S(m; t_d, w_d) (m - m_{0,d})^{a_d} \exp \left[-\left(\frac{m - m_{0,d}}{\lambda_d} \right)^{p_d} \right]$$

2021 functional form **fShiftSigPowTail** (excluding 65 MeV region)

$$B_d(m) = N_d S(m; t_d, w_d) (m - m_{0,d})^{a_d} \exp \left[-\frac{m - m_{0,d}}{\theta_d} + c_{1,d} u_d + c_{2,d} u_d^2 \right]$$

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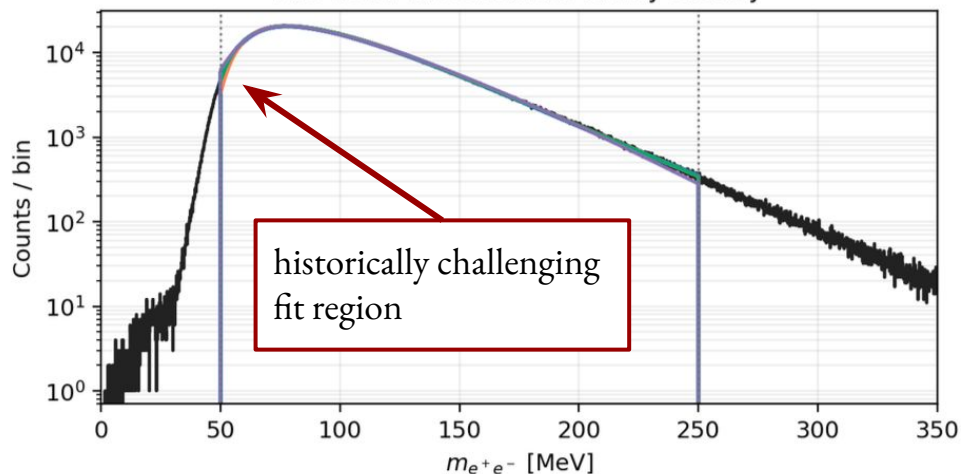
Four different statistical validation tests conducted

- I. 1% Fit * 10 (to double check 10% decisions)
- II. 1% Fit * 100 (to predict 100%)
- III. 10% Fit (to validate modeling on 10% data compare with scaled 1%)
- IV. 10% Fit * 10 (to predict 100% and compare with scaled 1%)

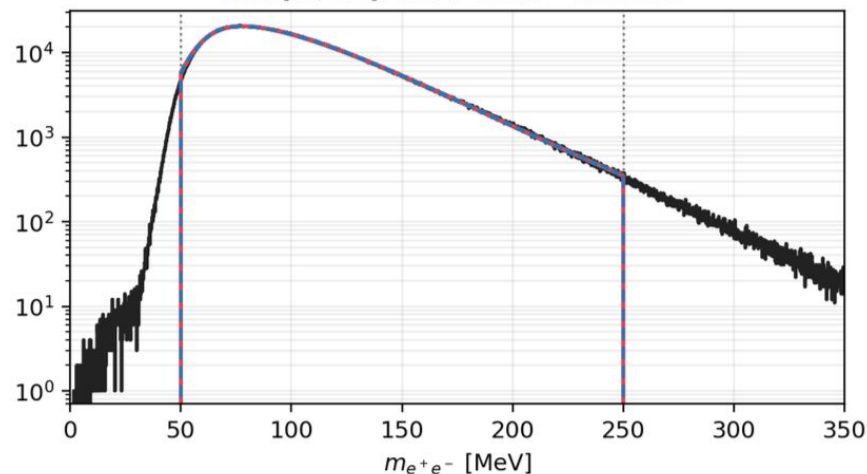
Example Fits to 1% Dataset

- Input histogram
- sigmoid*power*exp + raw expquad ($\chi^2/\text{ndof}=1.17$)
- thresholded gen-gamma ($\chi^2/\text{ndof}=1.07$)
- shifted gen-gamma ($\chi^2/\text{ndof}=1.19$)
- endpoint-aware sigmoid*power*exp ($\chi^2/\text{ndof}=3.75$)

HPS 2021 1% candidate-family overlay



HPS 2021 1% / sigmoid*power*exp + raw expquad
Scan [50, 250] MeV / N events = 1.229×10^7



Validation metrics

$$p_t(z) = \frac{\hat{A}_t(z) - A_{\text{inj}, t}(z)}{\sigma_{A, t}(z)}$$

$\hat{A}$: fitted yield; A_{inj} : injected yield.

σ_A : this toy's post-injection fit error.

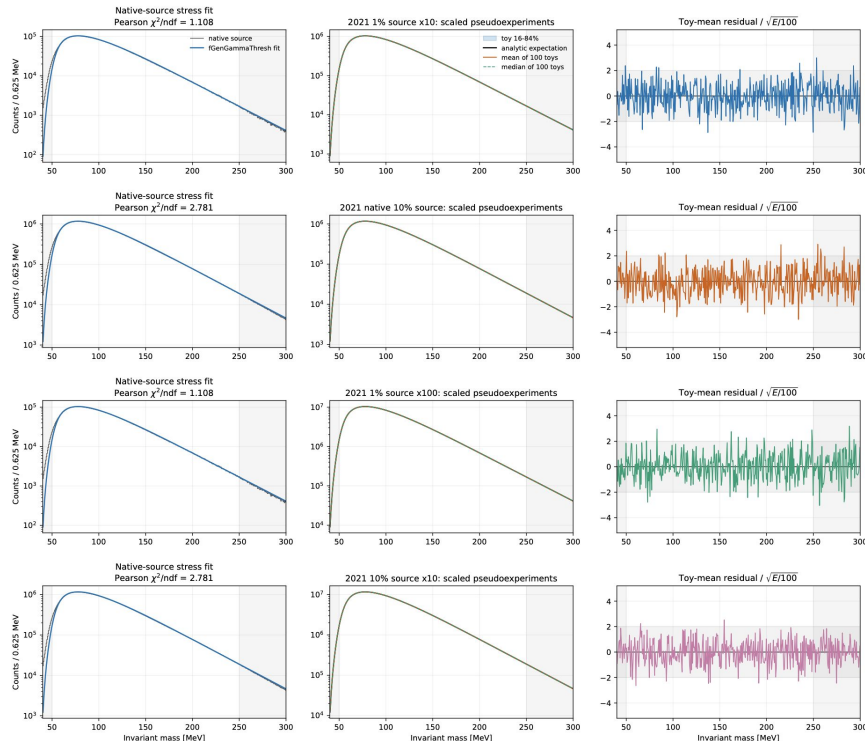
Pull mean $\rightarrow$ yield bias

Target: mean near 0.

Pull width $\rightarrow$ uncertainty scale

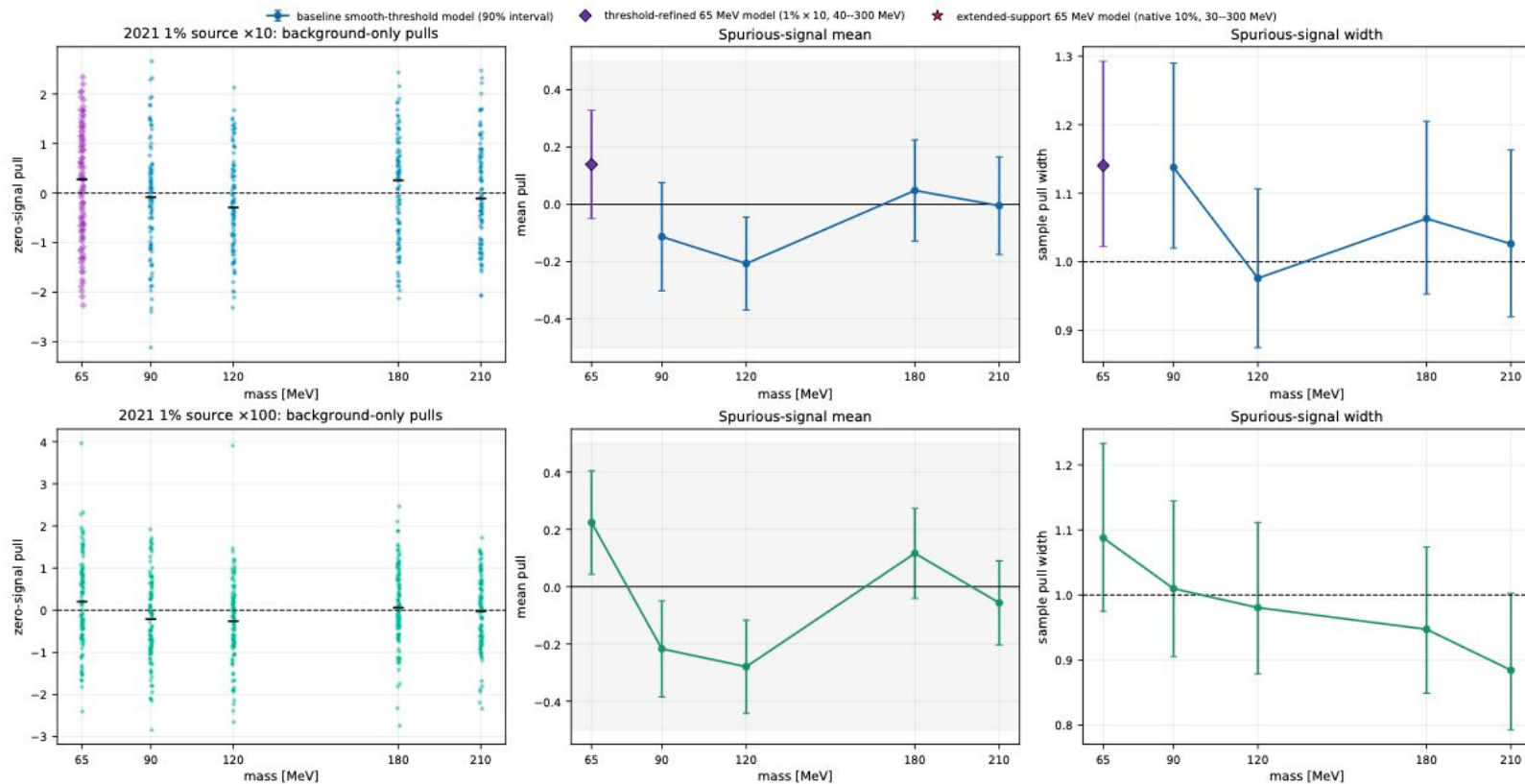
Target: width near 1.

Test zero-signal toys and 1, 3, 5 σ injections.

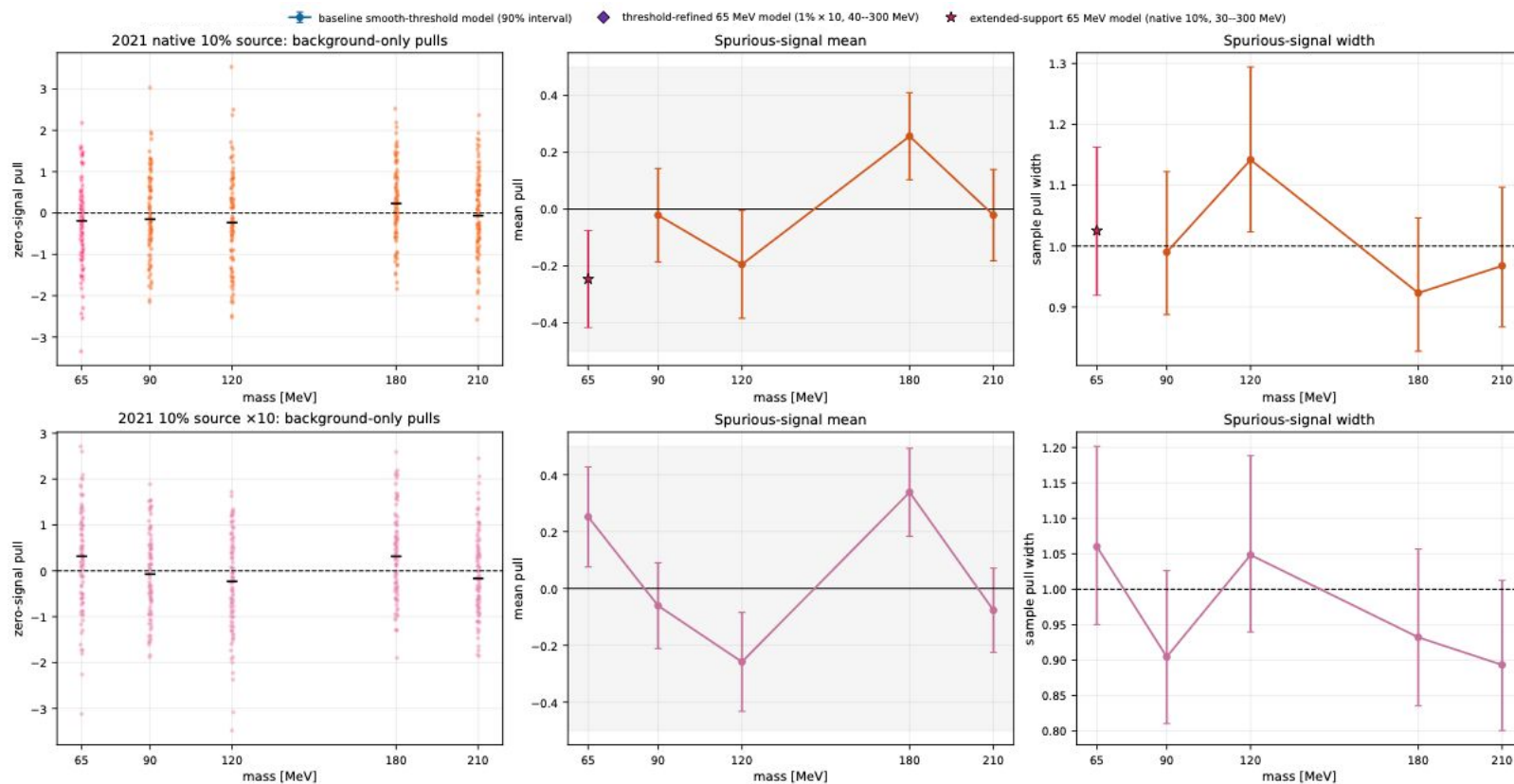


Conditional on the tested source; this is not a coverage result.

BKG Only Pseudoexperiments (1% Source)

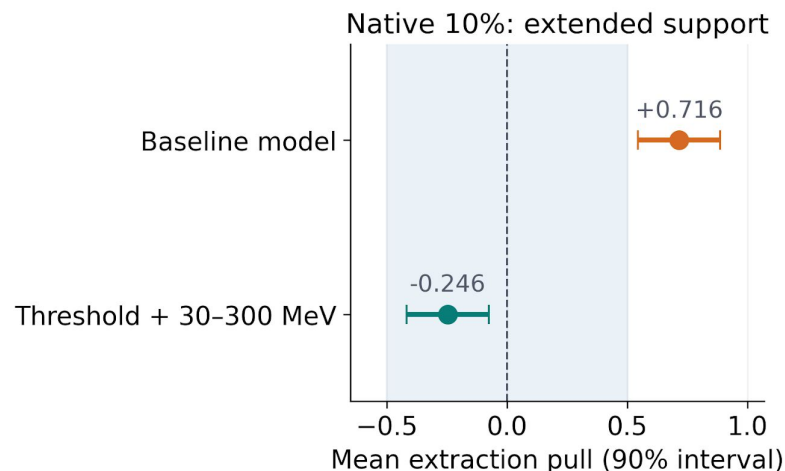
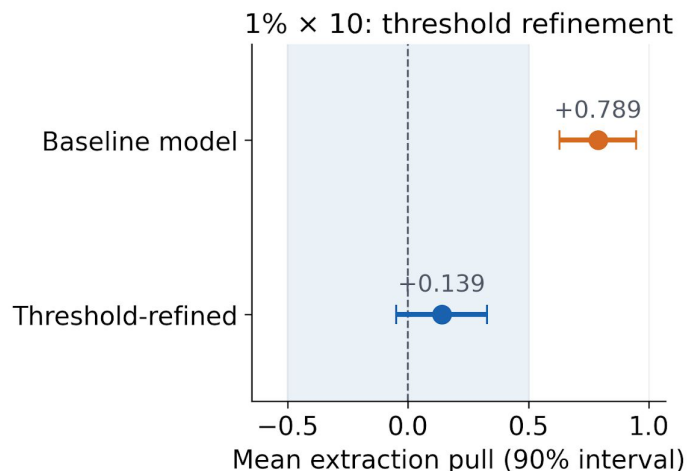


BKG Only Pseudoexperiments (10% Source)



Explanation of threshold modeling and alternate form

- 1% × 10: scale the fitted 1% mean; degree-five threshold continuation, with 40–300 MeV GP support.
- Native 10%: fit the saved 10% source; degree-six threshold continuation and 30–300 MeV support.



65 MeV, 100 backgrounds per ensemble. Shading: post-result practical ± 0.5 band; not a coverage test.

Set the yield from a matched reference fit

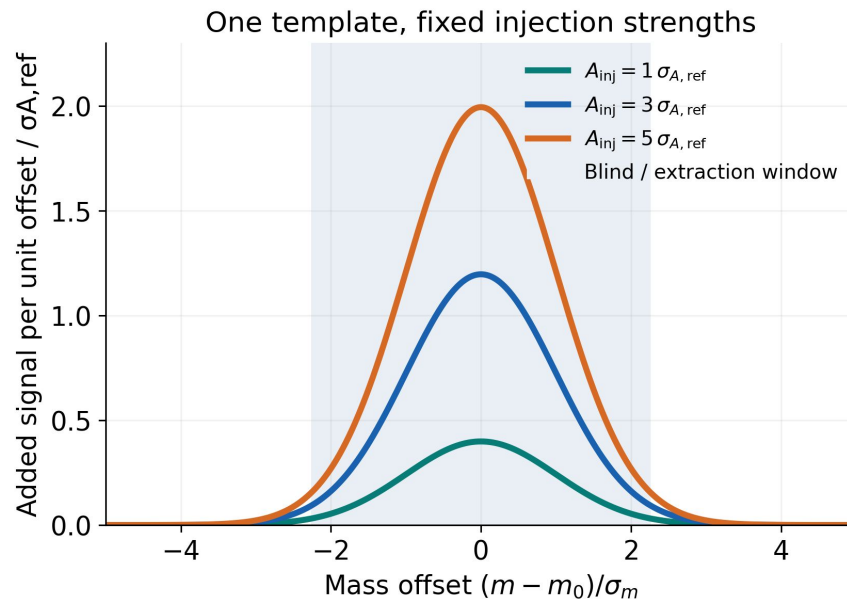
$$A_{\text{inj}}(z) = z \sigma_{A, \text{ref}}, \quad z \in \{0, 1, 3, 5\}$$

Use the background-only fit error for this source, toy and mass.

Add full Gaussian templates to matched background toys.

Refit the sidebands, then repeat the signed signal extraction.

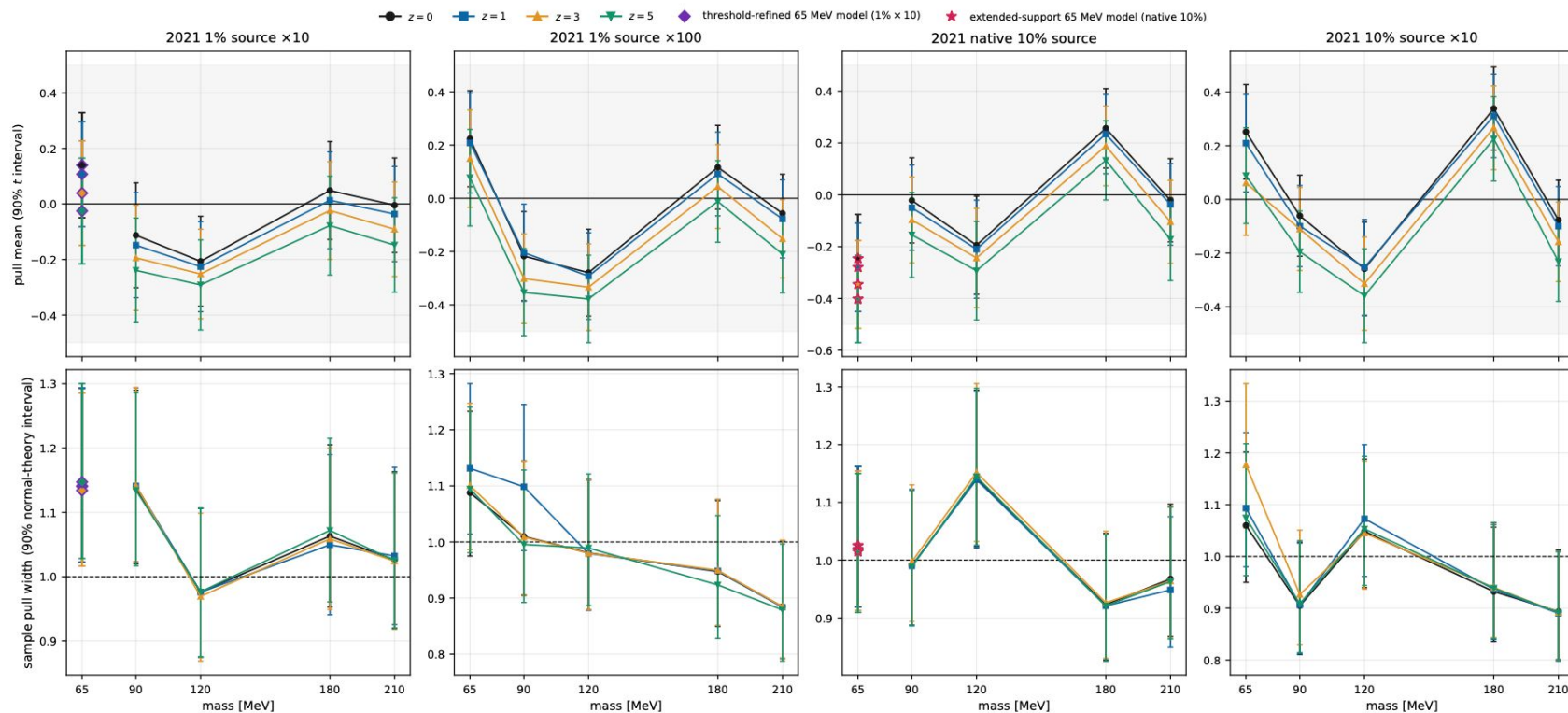
The reference error sets the injected yield; it does not guarantee the measured significance.



Illustration; area under each curve gives its injected yield.

Injected Signal Studies

Signal-injection pull diagnostics across 100 pseudoexperiments per point



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Bounded profile-likelihood ratio (non-negative signal)

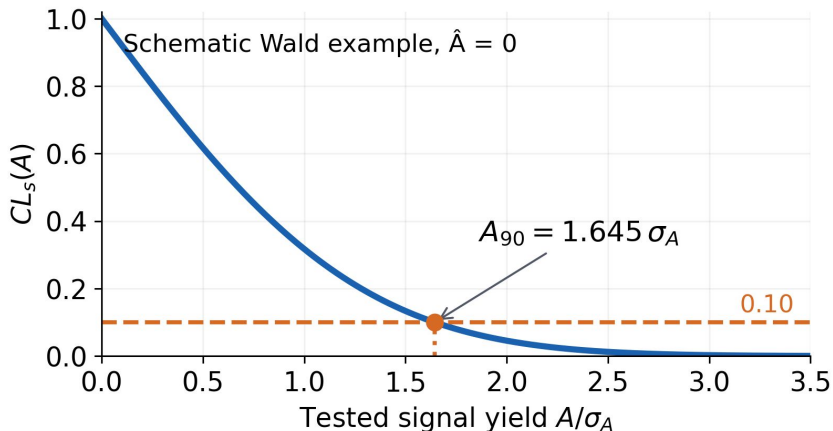
$$q_{0,d}(m) = -2 \ln \frac{\mathcal{L}_d\left(A_d = 0, \hat{\hat{\theta}}_d(0)\right)}{\mathcal{L}_d\left(\hat{A}_d, \hat{\theta}_d\right)}$$

Profile the background in both hypotheses.

An excess gives $q_0 > 0$; a deficit is assigned $q_0 = 0$.

Observed Upper Limits

1. Test a signal yield A at a fixed mass.
2. Profile the background for each A .
3. The 90% limit is where CLs reaches 0.10.



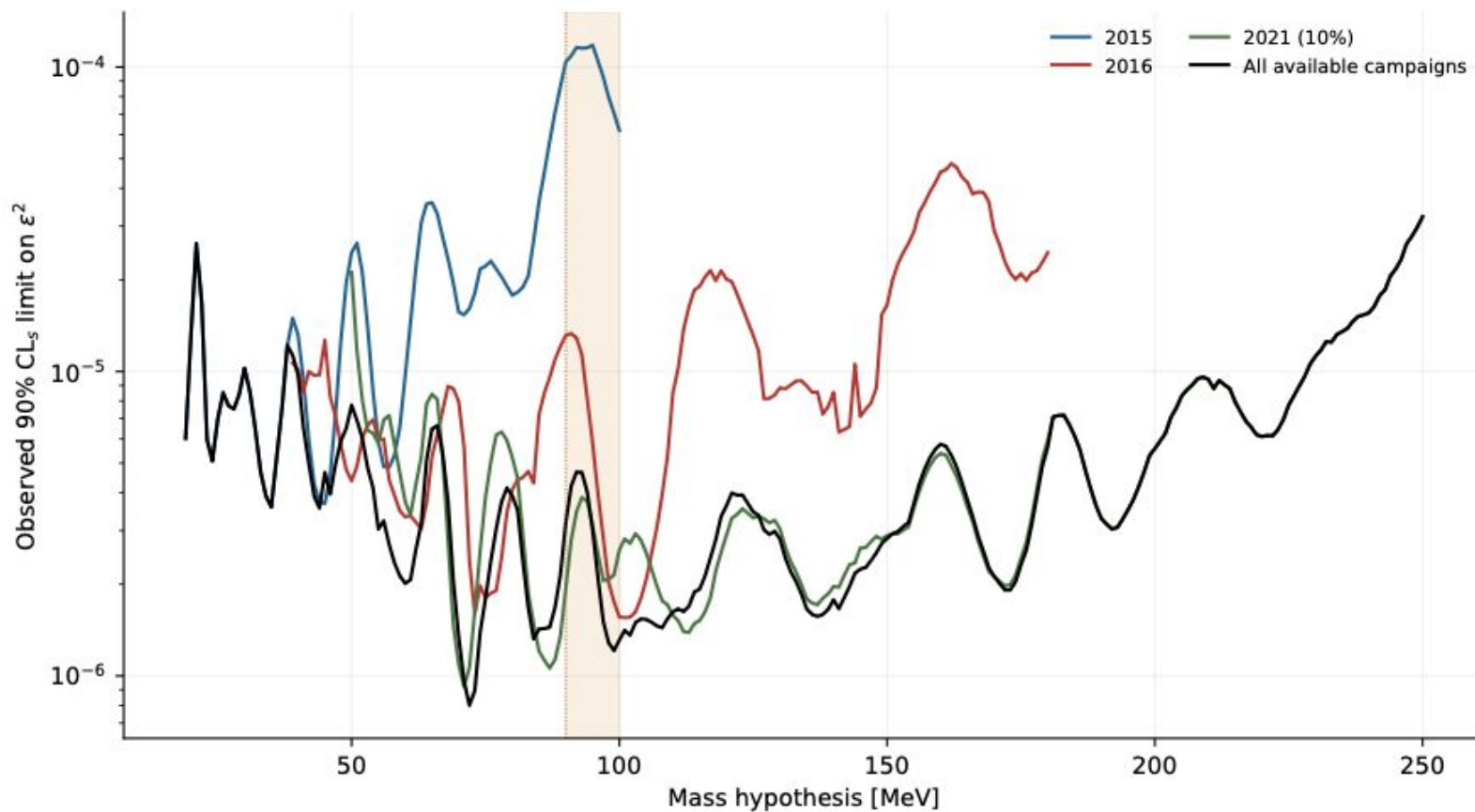
$$\tilde{q}_{A_d}(m) = \begin{cases} 0, & \hat{A}_d^{\text{unc}} > A_d, \\ -2 \ln \frac{\mathcal{L}_d(A_d, \hat{\theta}_d(A_d))}{\mathcal{L}_d(\hat{A}_d^{\text{unc}}, \hat{\theta}_d)}, & 0 \leq \hat{A}_d^{\text{unc}} \leq A_d, \\ -2 \ln \frac{\mathcal{L}_d(A_d, \hat{\theta}_d(A_d))}{\mathcal{L}_d(0, \hat{\theta}_d(0))}, & \hat{A}_d^{\text{unc}} < 0, \end{cases}$$

$$CL_b(A_d; m) = P(\tilde{q}_{A_d} \geq \tilde{q}_{A_d}^{\text{obs}} | \text{bkg})$$

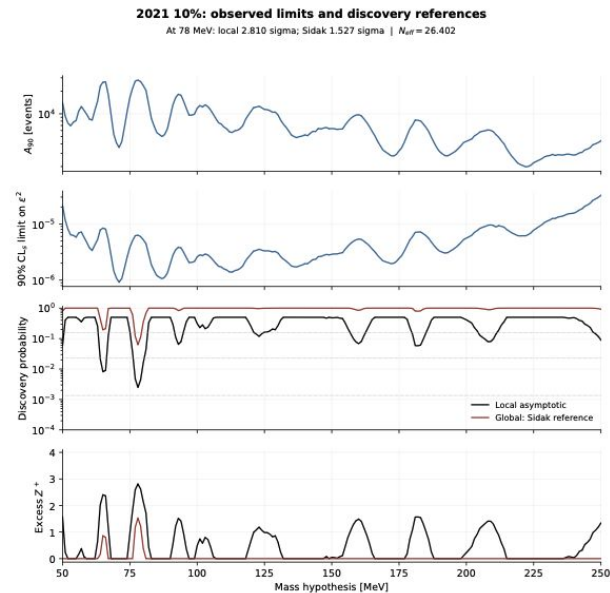
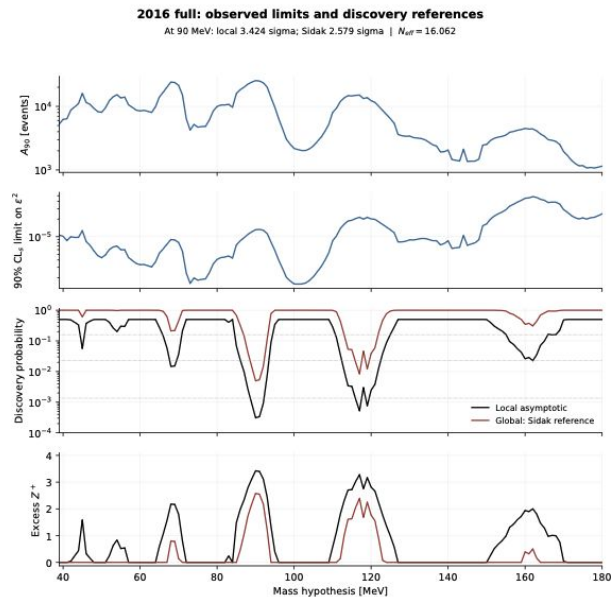
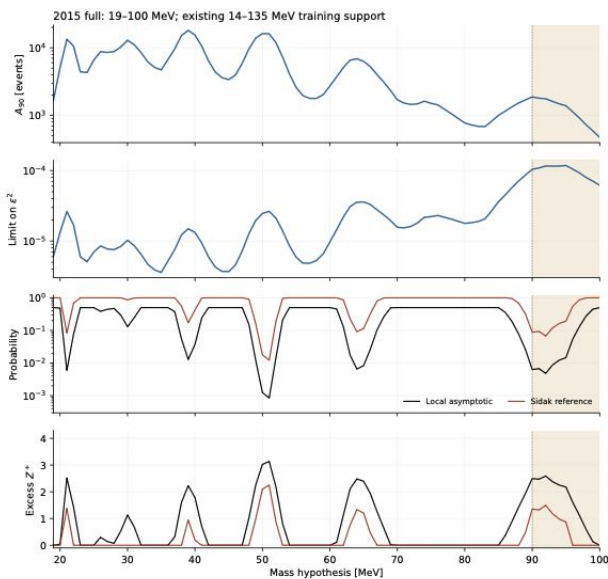
$$CL_{s+b}(A_d; m) = P(\tilde{q}_{A_d} \geq \tilde{q}_{A_d}^{\text{obs}} | A_d + \text{bkg})$$

$$CL_s(A_d; m) = \frac{CL_{s+b}(A_d; m)}{CL_b(A_d; m)}$$

Observed Upper Limits



Observed Limits and Discovery Pvalues for individual datasets

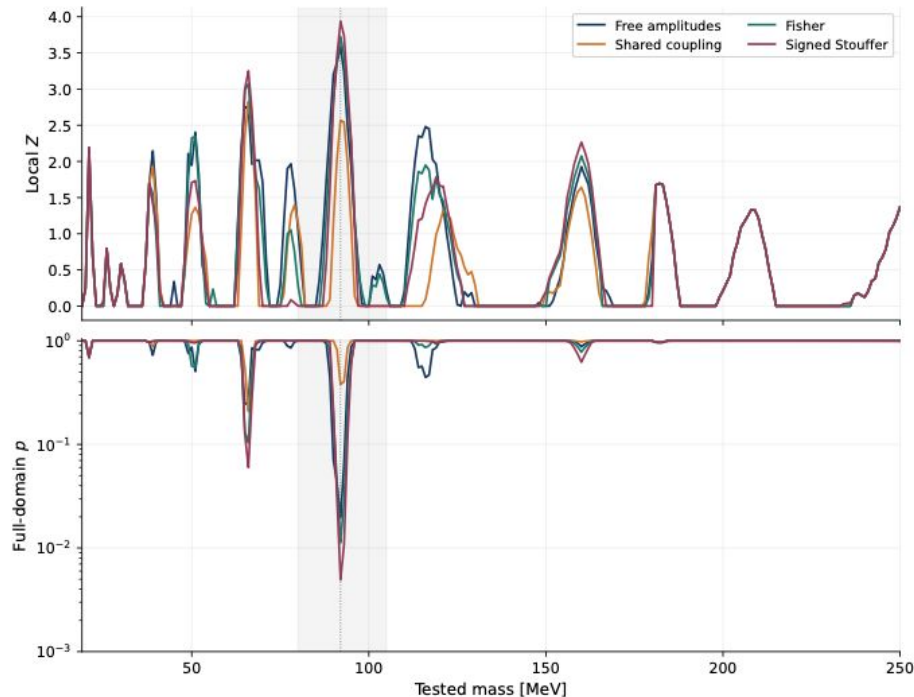


Combined discovery significance

Combined Significance dependent
on underlying phenomenological
framework

Shared Coupling: 2.8 sigma

Individual Experiments: 3.9 sigma

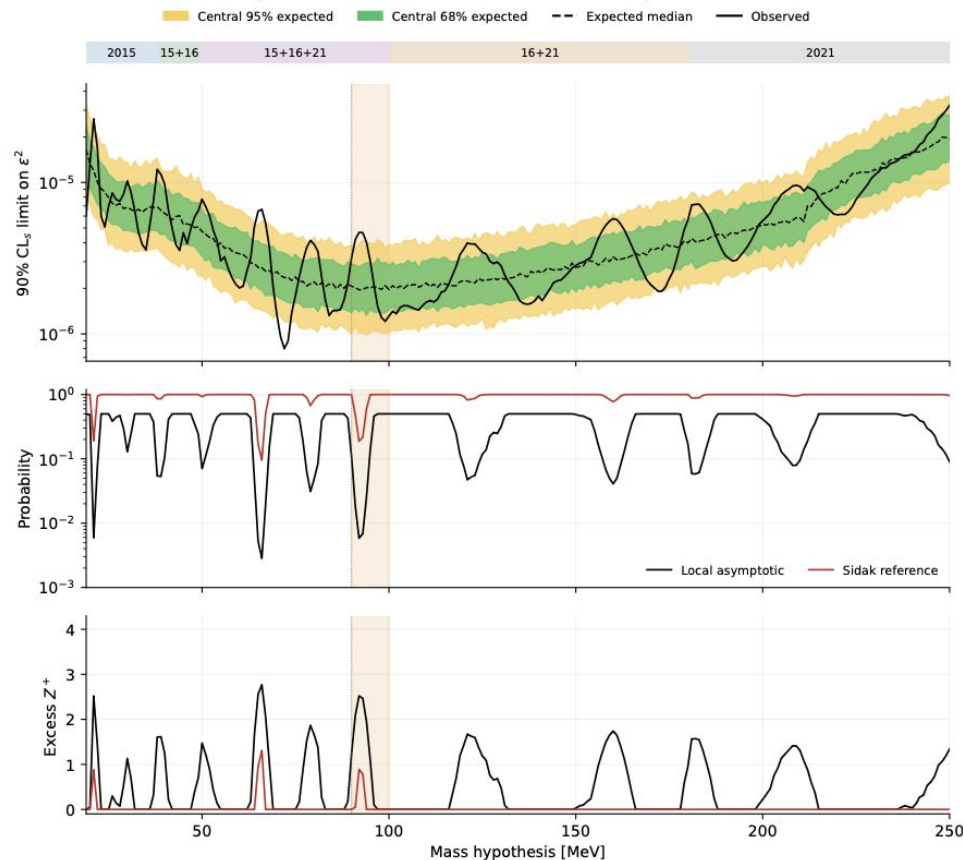


Combined Search Results

300 Toys for bands

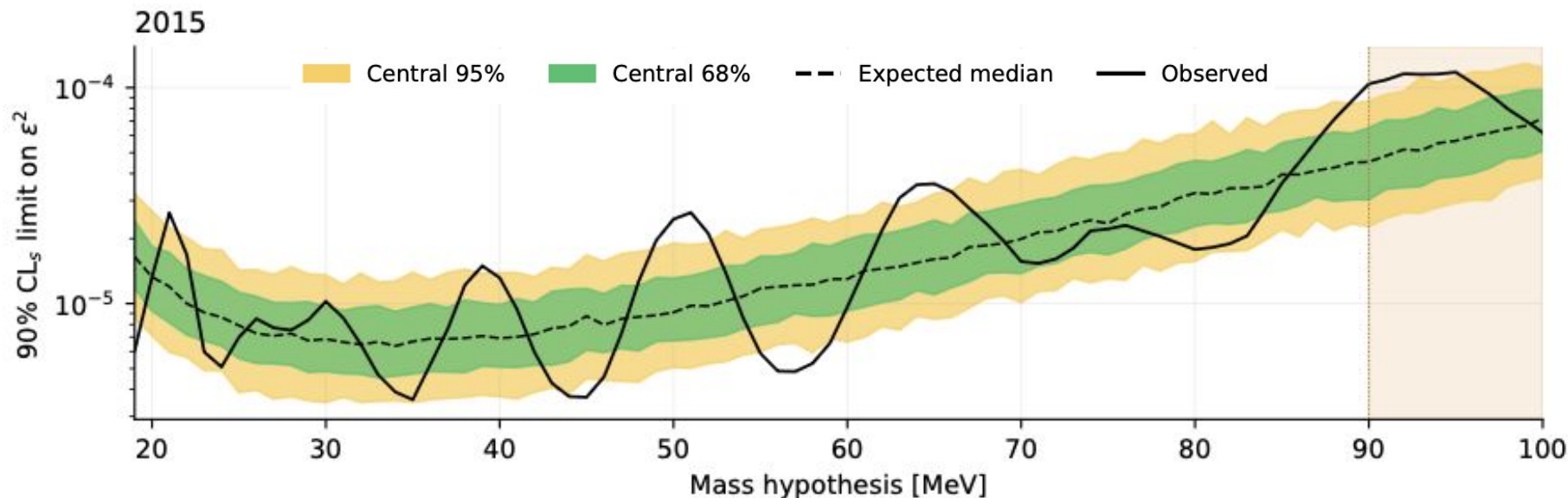
LEE penalty factor of 35

- Most significant local 2.8
- Most significant local 1.4

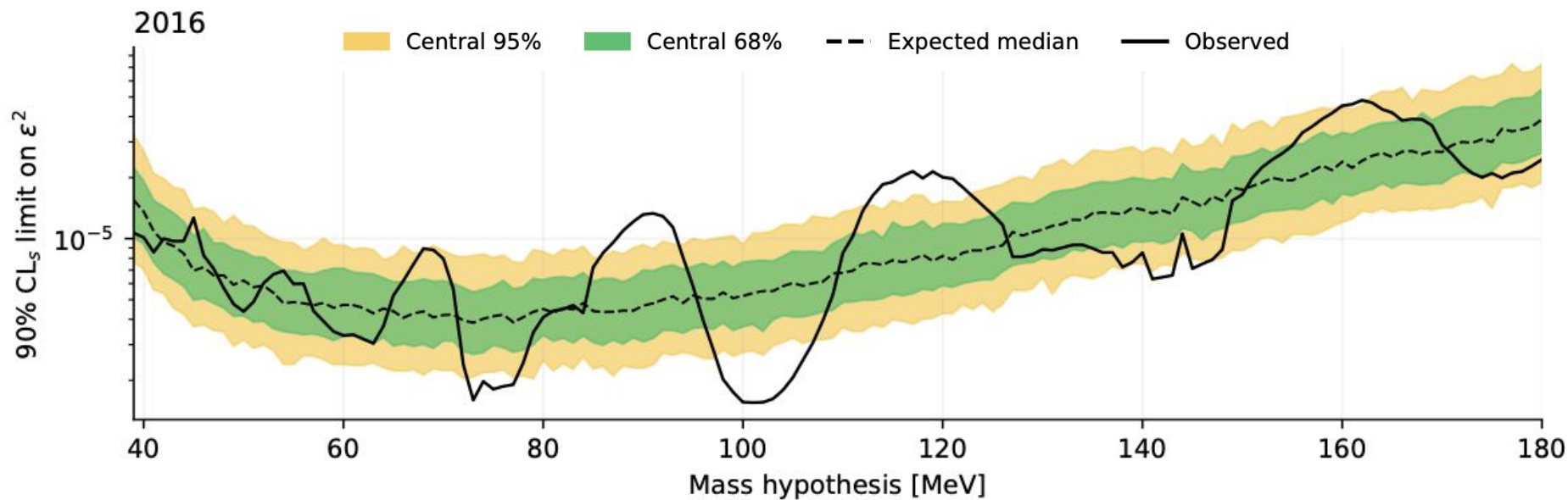


2015 Individual Limits

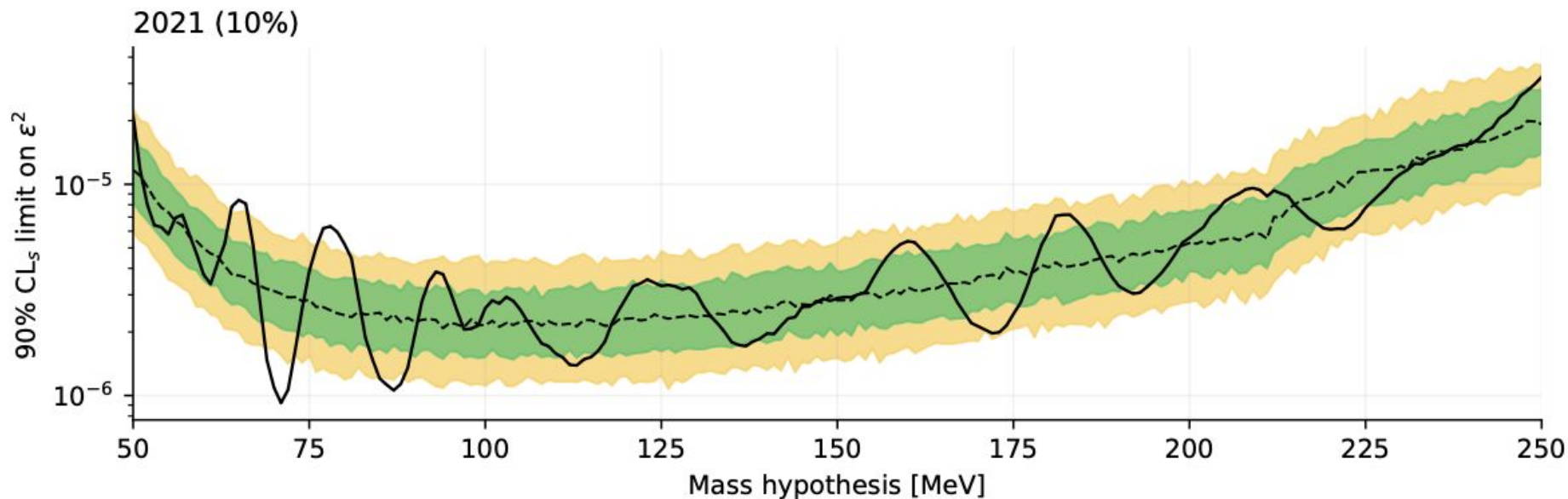
Comparison to re-analysis in backup



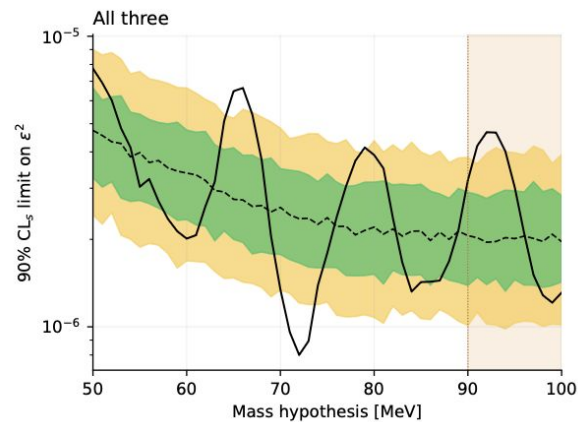
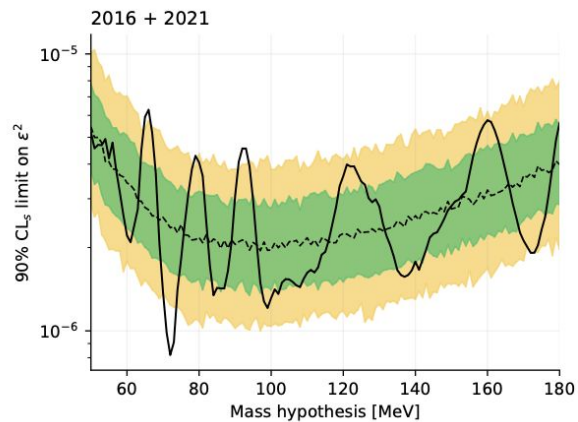
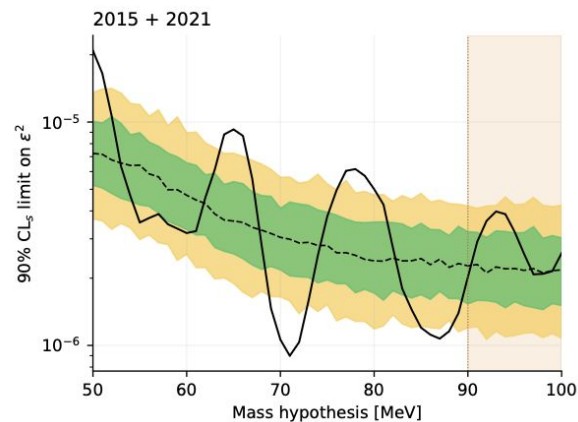
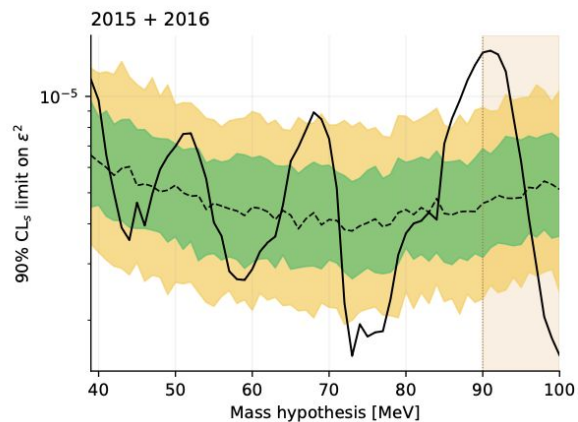
2016 Individual Limits



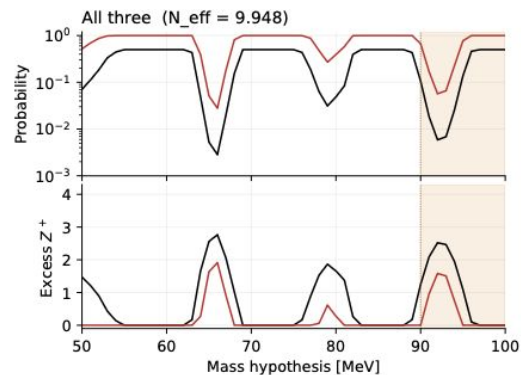
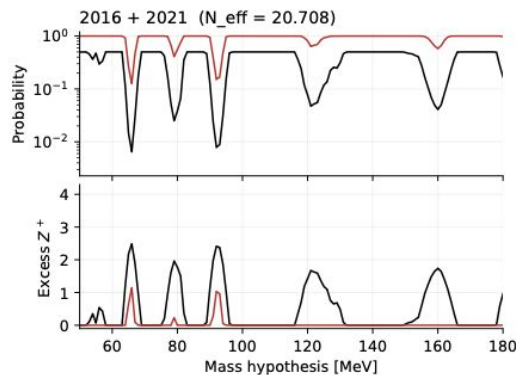
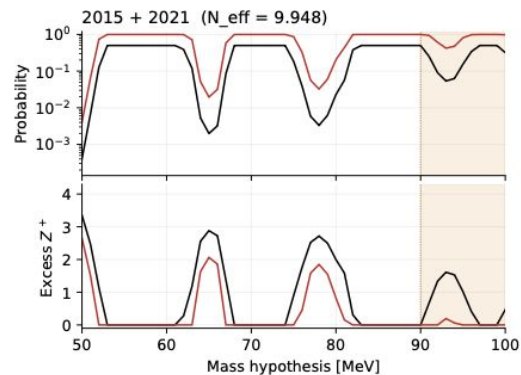
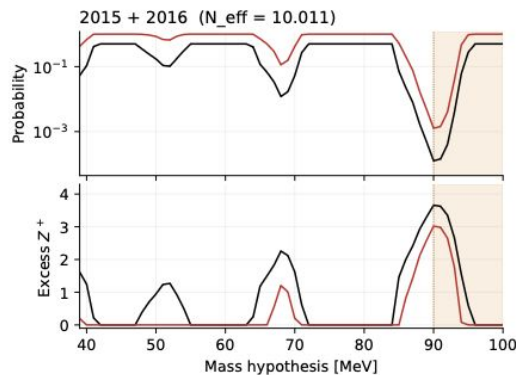
2021 10%



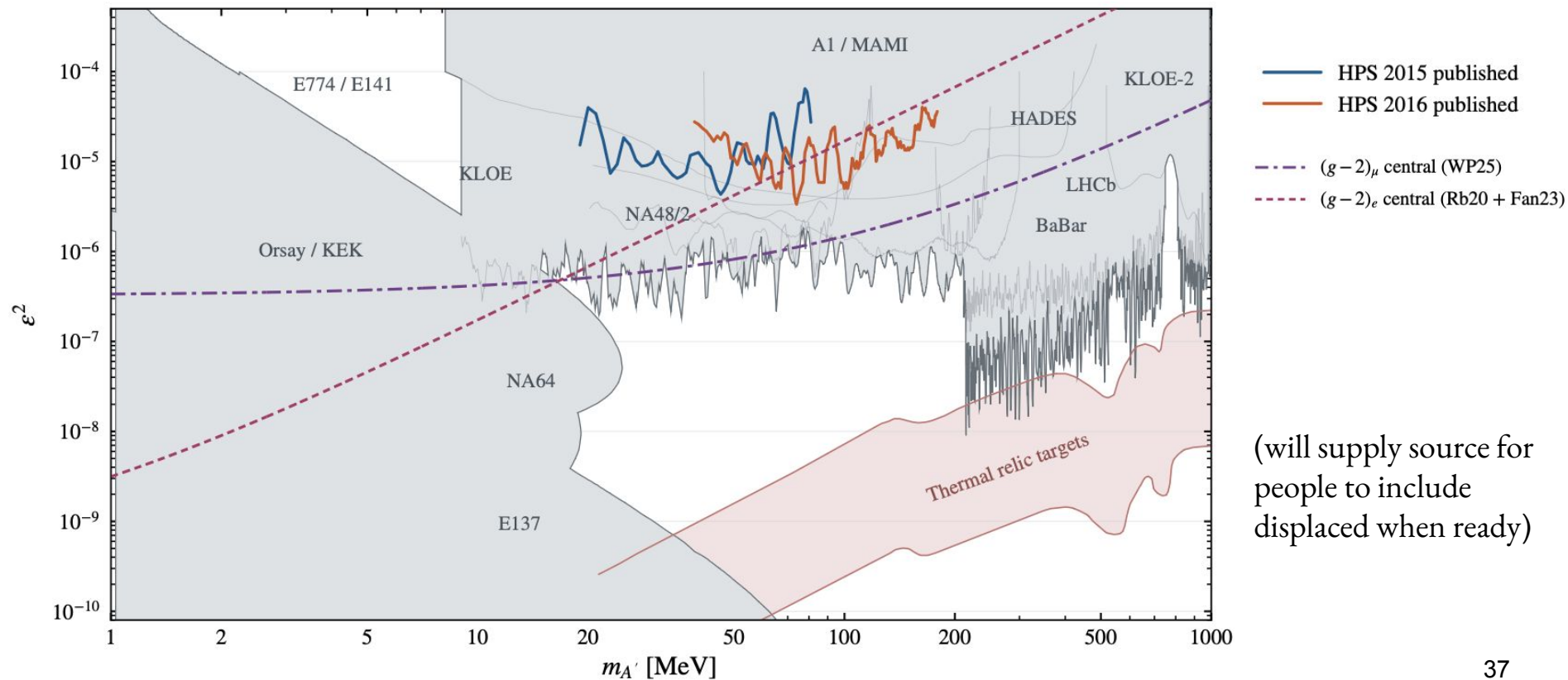
All Combined Regions



Combined region discovery pvalues



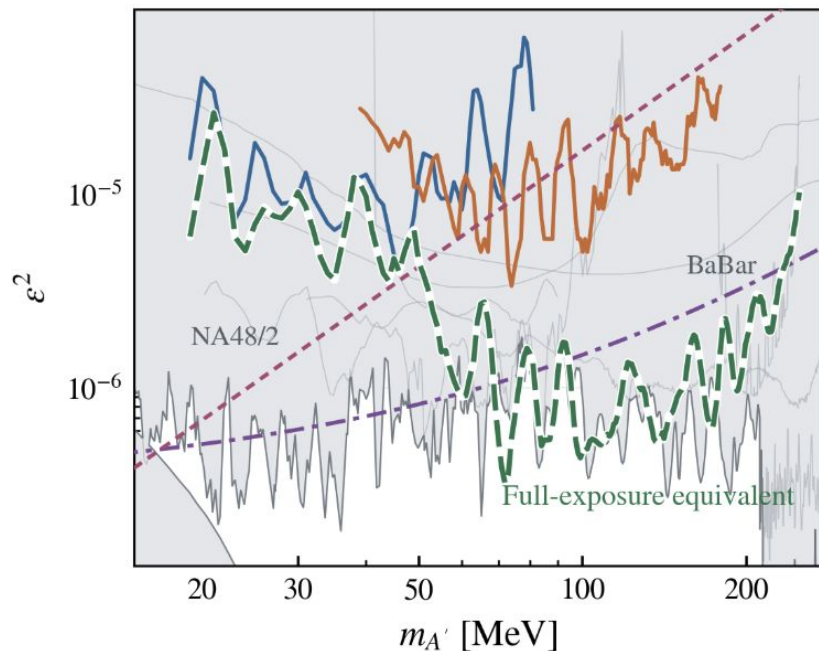
Making Cleaner and Modern Contours



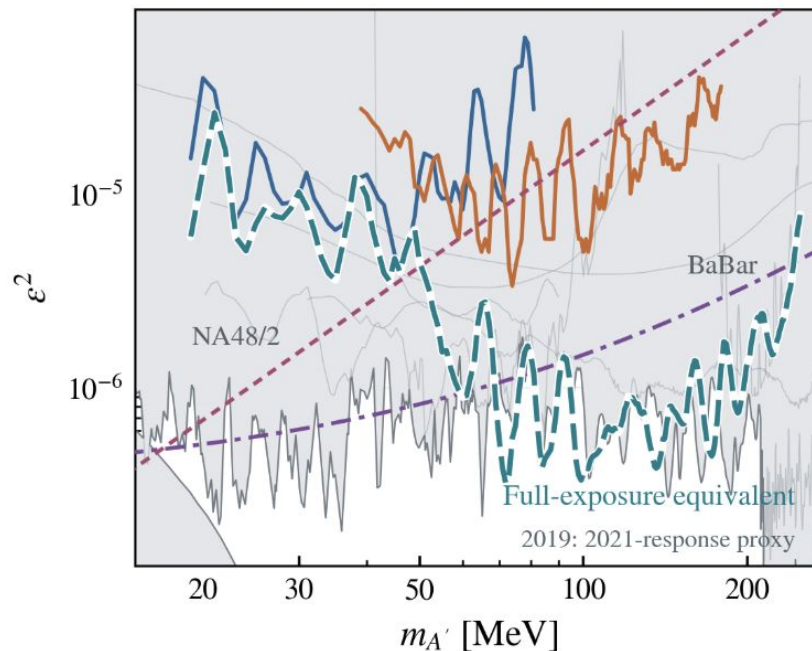
(will supply source for people to include displaced when ready)

Projected Contours

(a) 2015 + 2016 + 2021



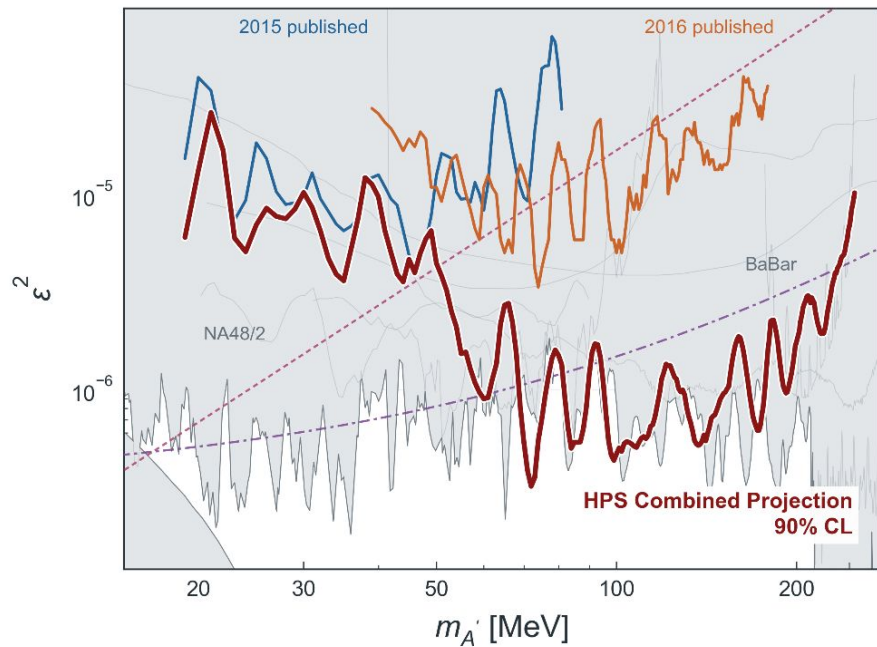
(b) 2015 + 2016 + 2019 + 2021



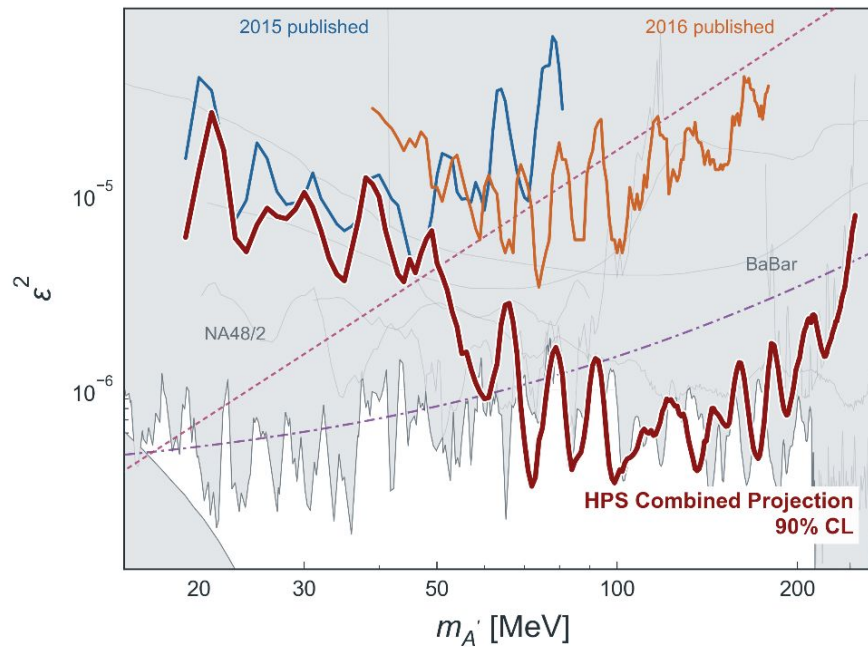
More thorough babar comparison at end of presentation.

2019 comes from 1% UC sample, used for luminosity equivalent and ~coverage only.

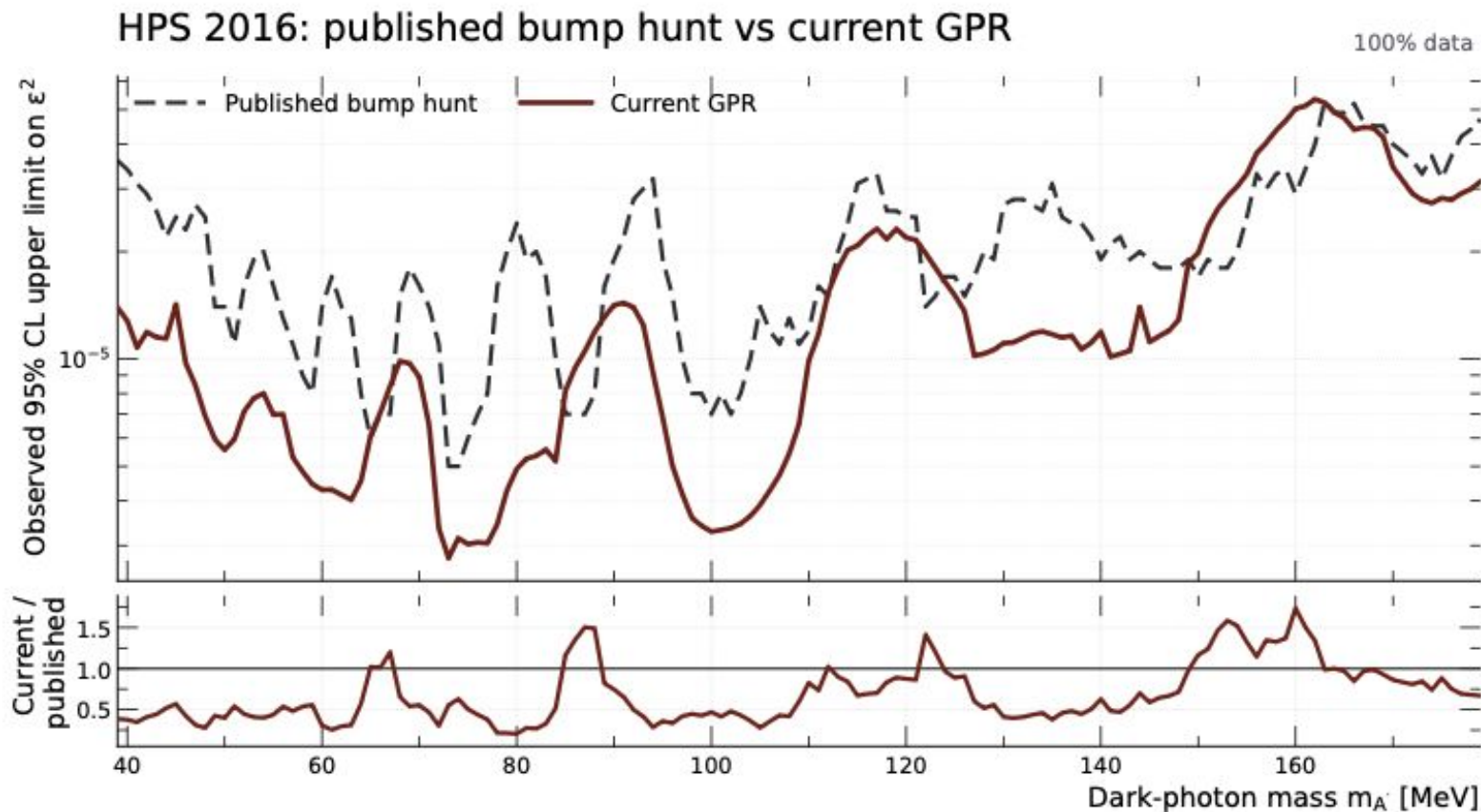
(a) 2015 + 2016 + 2021



(b) 2015 + 2016 + 2019 + 2021



Observed Limit Comparison with 2016 Publication



What would a signal look like in full 2021 dataset?

- must be able to map correlations if they exist in invariant mass
- must be able to characterize shape of observed upper limit with signal injected

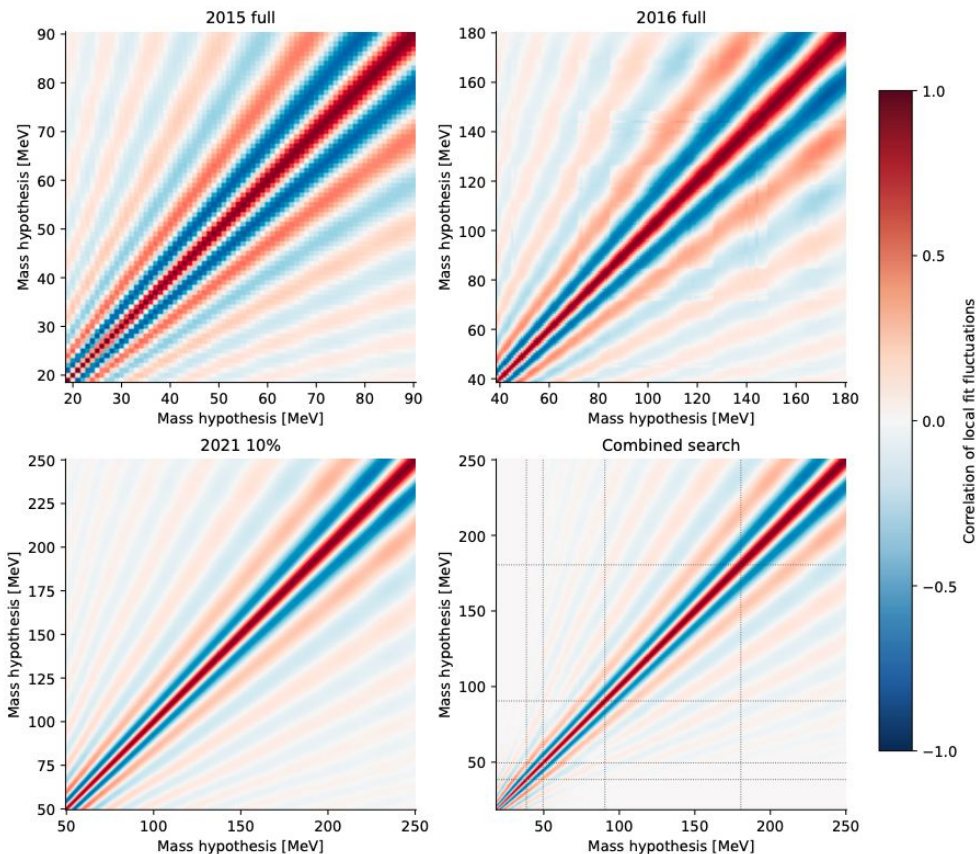
What is the max significance a signal could be in 2021?

GPR Correlations in mass ($1/n$)

This mapping shows that for a local 1 sigma fluctuation there is a strong anti-correlation nearby that oscillates as you move away

Makes sense!

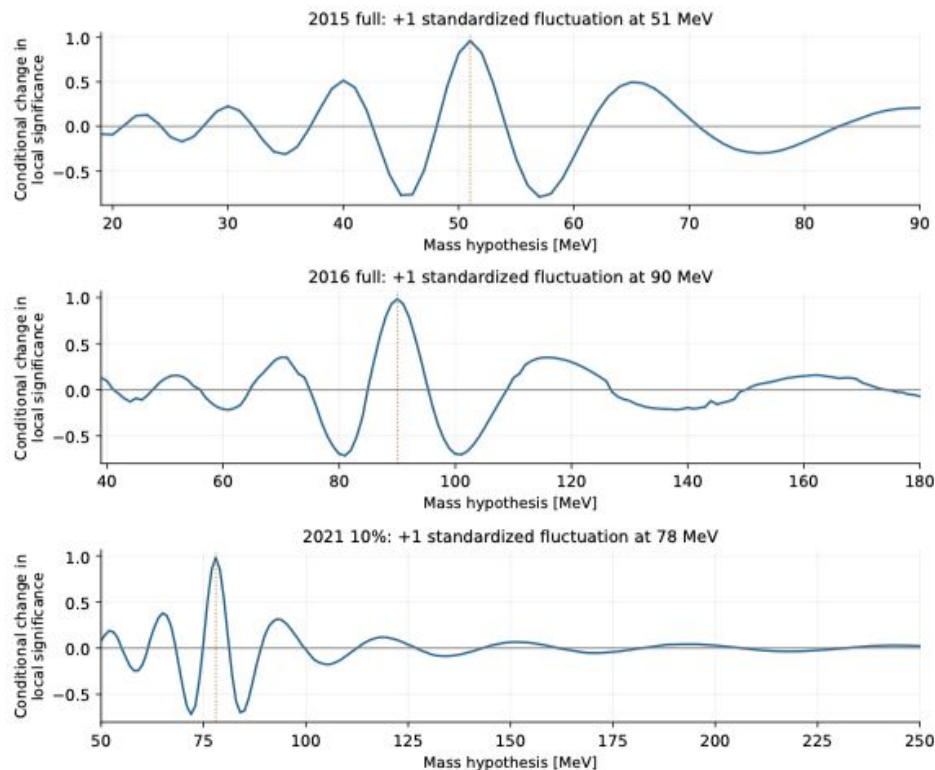
Signal in a training sideband will of course spoil the training nearby



GPR Correlations in mass ($1/n$)

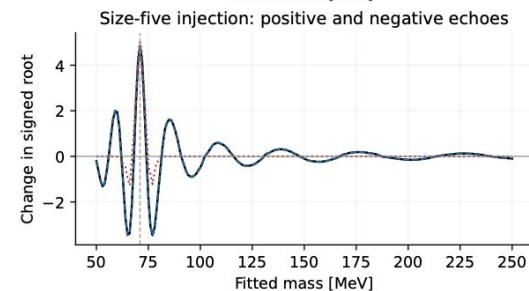
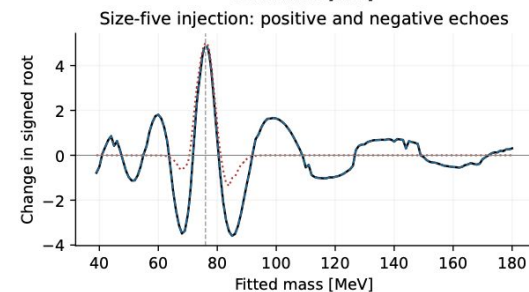
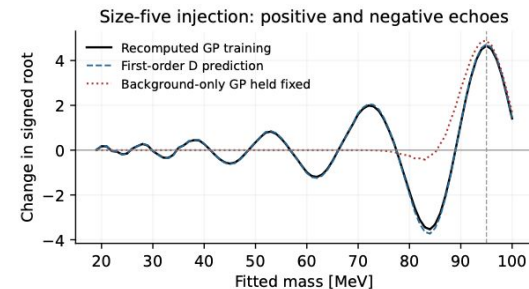
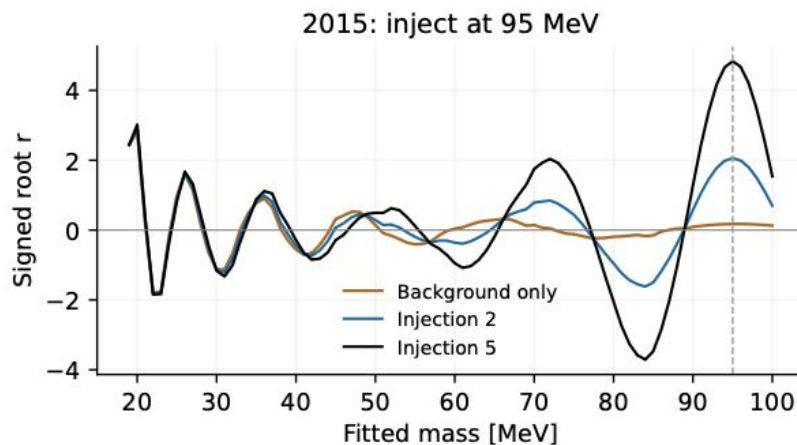
Size/shape of correlated
“echoes” dependent on

- mass resolution
- location of peak
- underlying background of nearby regions



First order prediction for location of features

$$\Delta r_m^{\text{lin}} = \sum_{i: B_i > 0} D_{im} \frac{\delta n_i}{\sqrt{B_i}}$$



Projection of max signal if regions of interest are true signal

$$S_{100\%} = kS_f, \quad B_{100\%} = kB_f, \quad Z_{100\%} \simeq \frac{kS_f}{\sqrt{kB_f}} = \sqrt{k} Z_f$$

$$Z_{1\% \rightarrow 100\%}^{\text{target}} = 10Z_{1\%}$$

$$Z_{10\% \rightarrow 100\%}^{\text{target}} = \sqrt{10} Z_{10\%}.$$

Conditional persistence test: one region at a time

- Scale the fitted 10% continuum to full exposure ($k = 10$).
- Choose a signal yield whose Asimov fit reaches $\sqrt{10} Z_{10}\%$.
- Draw 20 Poisson spectra; rerun the moving-window GP and extraction.

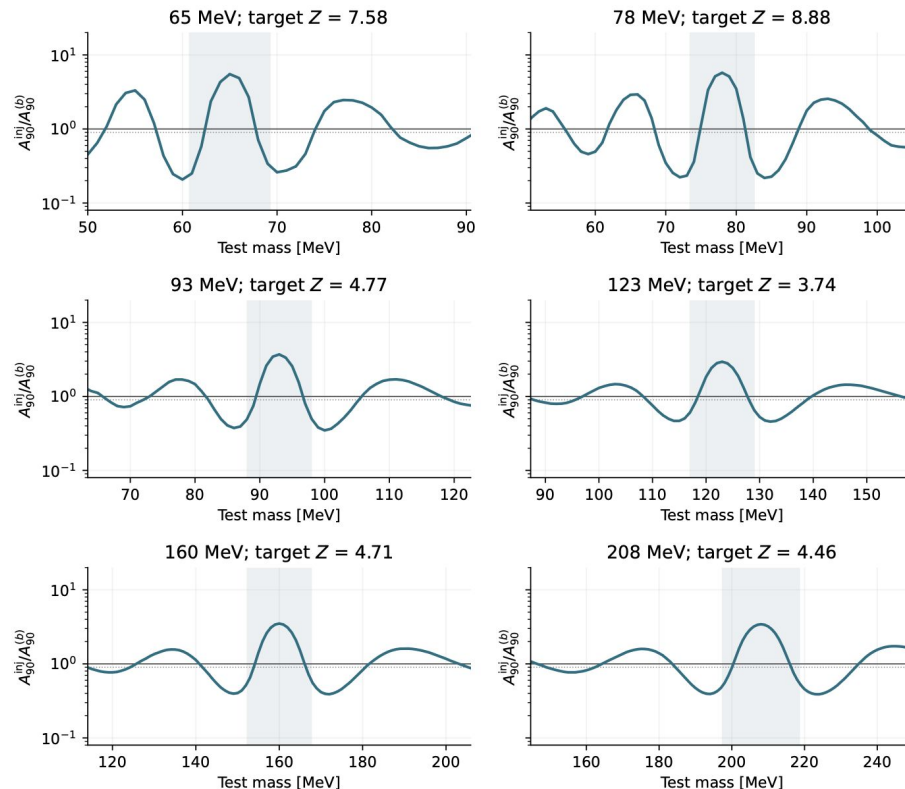
R compares injected and continuum limits. $R < 1$ means tighter neighboring limits; D measures the echo depth.

$$R(m) = \frac{A_{90}^{\text{injected}}(m)}{A_{90}^{(b)}(m)}, \quad D = 1 - R(m_{\text{echo}}), \quad m_{\text{echo}} \in m_0 \pm [2, 8] \sigma_m(m_0)$$

Predictions of upper limit structure

Have proven ability to predict
observed upper limit structure to
first order

projected shape of upper limits for
different regions of interest in full
2021 dataset



- I. Additional Local Likelihood using GP on projected significance
- II. 2021 Threshold fit optimization
- III. Signal Extraction Display
- IV. Mass Resolution Studies
- V. 2015 95% CLs Comparison
- VI. Projection to Babar Plot
- VII. Two Signal Hypothesis
- VIII. Potential fit catalogues
- IX. Interpretation slides

Local Likelihood using GP on projected significance: pt1

Model the signed-root scan response

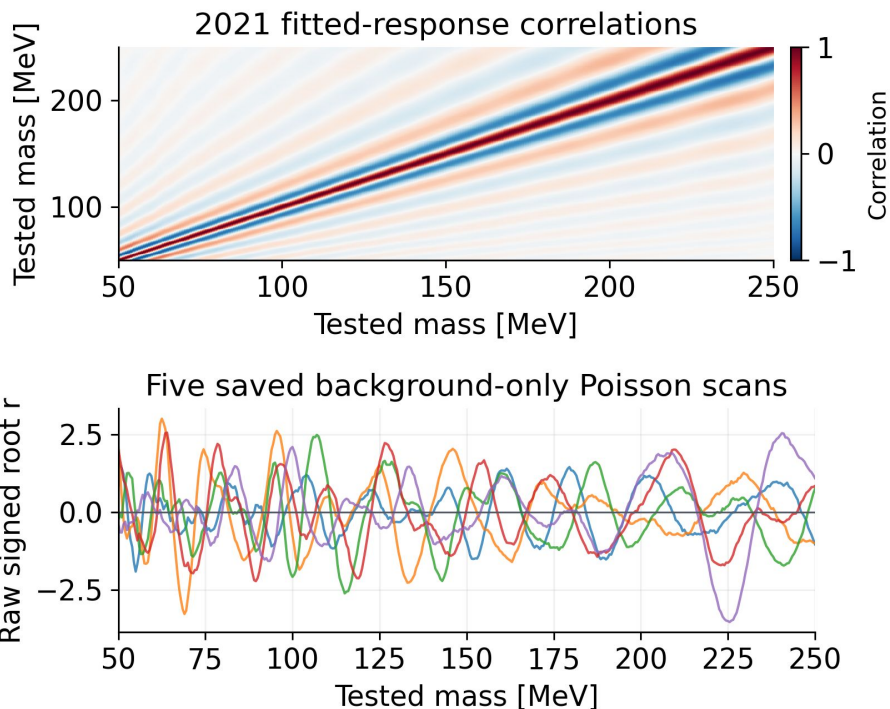
$$r^* \simeq a + D^T \xi, \quad \xi \sim N(0, I)$$

$$\Sigma_r = D^T D, \quad R_{ij} = \frac{\Sigma_{r, ij}}{s_i s_j}$$

D maps bin fluctuations into the signed-root mass scan.

Retain the response mean a , scale s and mass correlation R .

Draw correlated fields and record their largest excess.



Local Likelihood using GP on projected significance: pt2

One mass versus the full search

$$Z_{\text{local}} = \max(r_{\text{obs}}, 0)$$

$$T = \max_m \max(r(m), 0)$$

$$p_{\text{global}} = P_B(T^* \geq T_{\text{obs}})$$

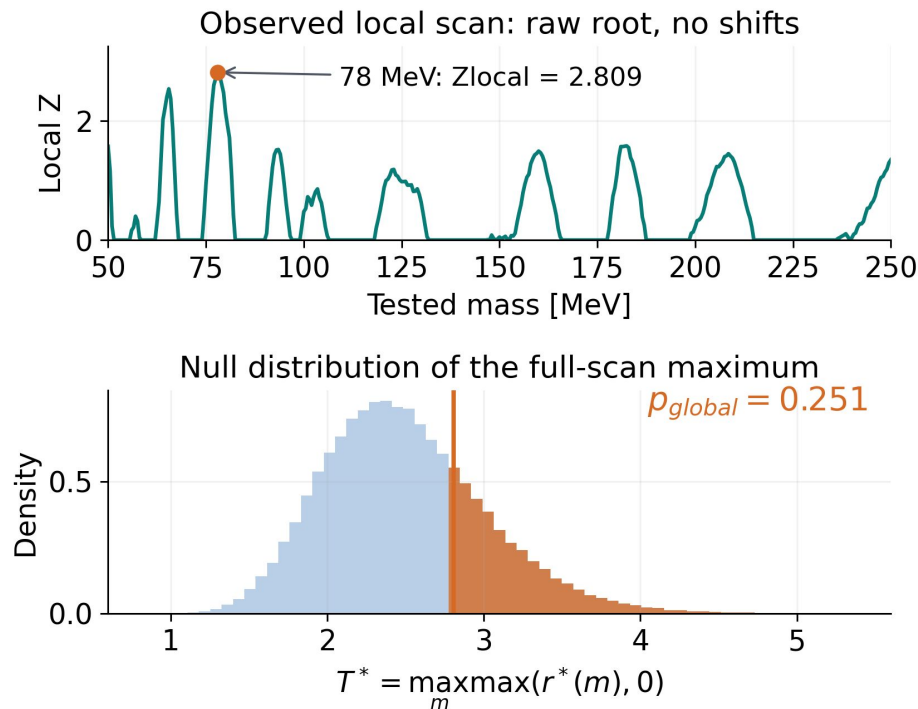
Local: the excess at one mass, with the raw asymptotic mapping.

Global: the largest excess anywhere in 50–250 MeV.

Poisson check: 57/256 exceedances.

95% interval: 0.173–0.279.

Ananiev & Read, JINST 18 (2023) P05041; HPS v5.8.5.3.



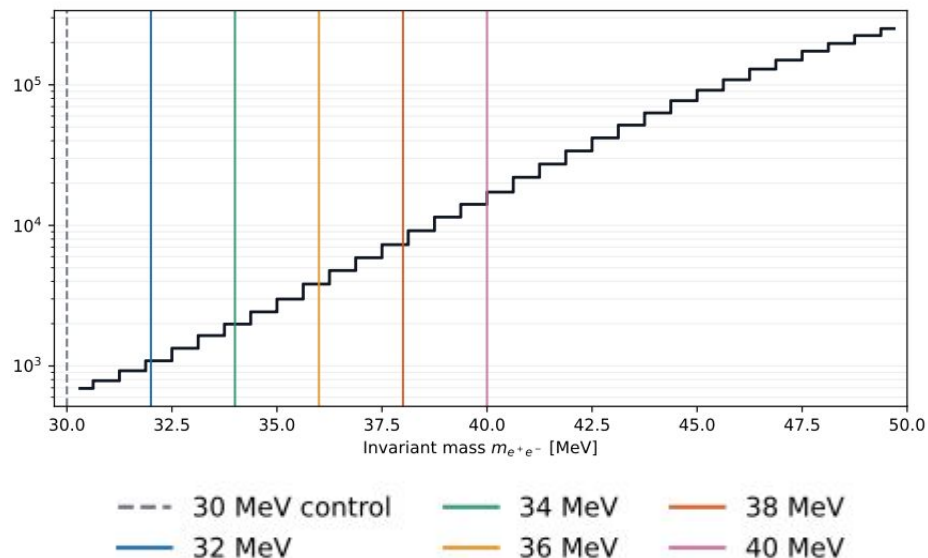
2021 Threshold Fit optimization

Study done after seeing apparent biases with GP fit support starting at 40 MeV in pseudoexperiment tests.

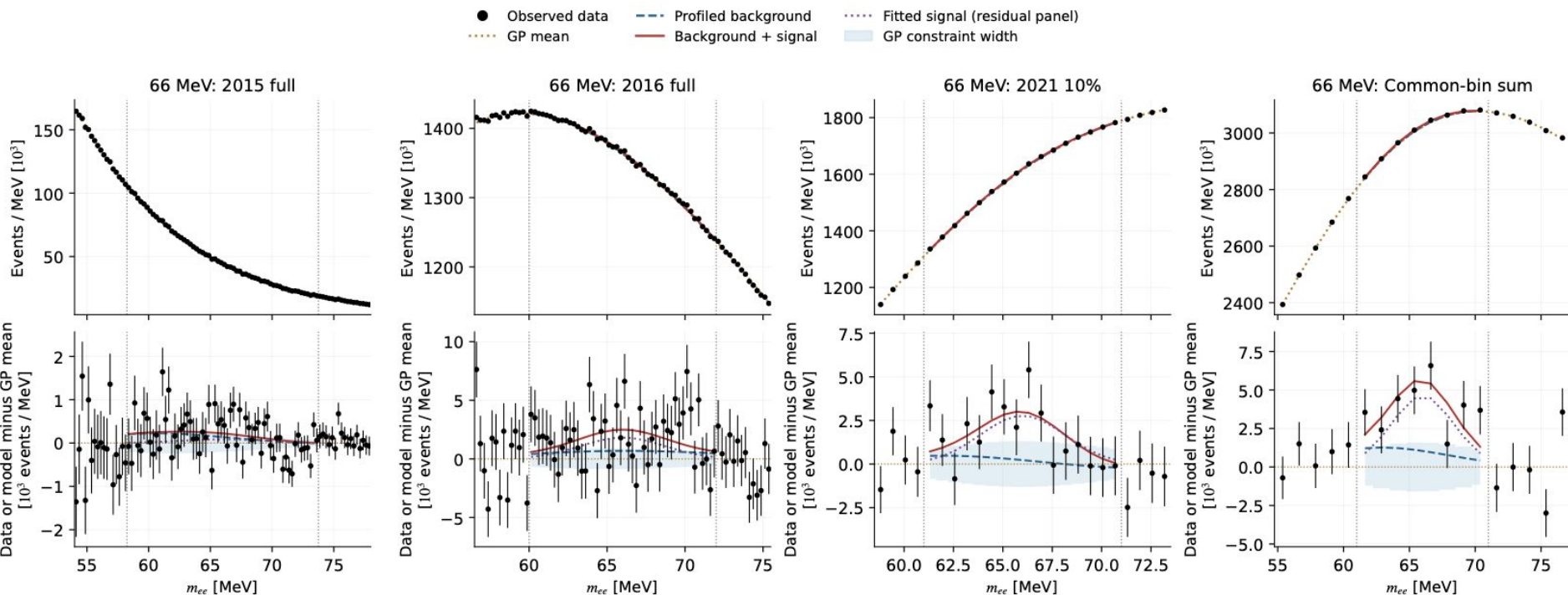
- selected GP fit support range and procedure for selection for fully unblinded 2021 dataset

Rising edge support selected: 36 MeV
Note goes into more detail

10% 2021 Threshold Region



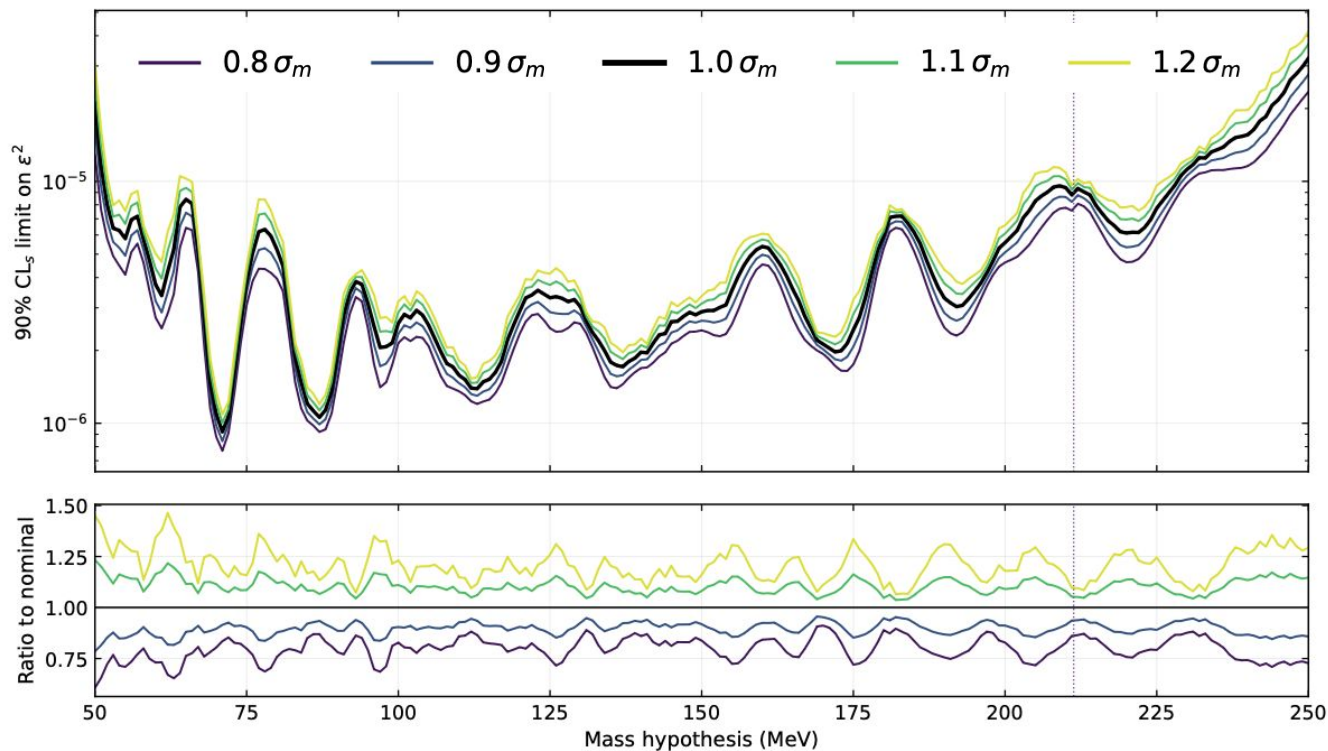
Extraction Display



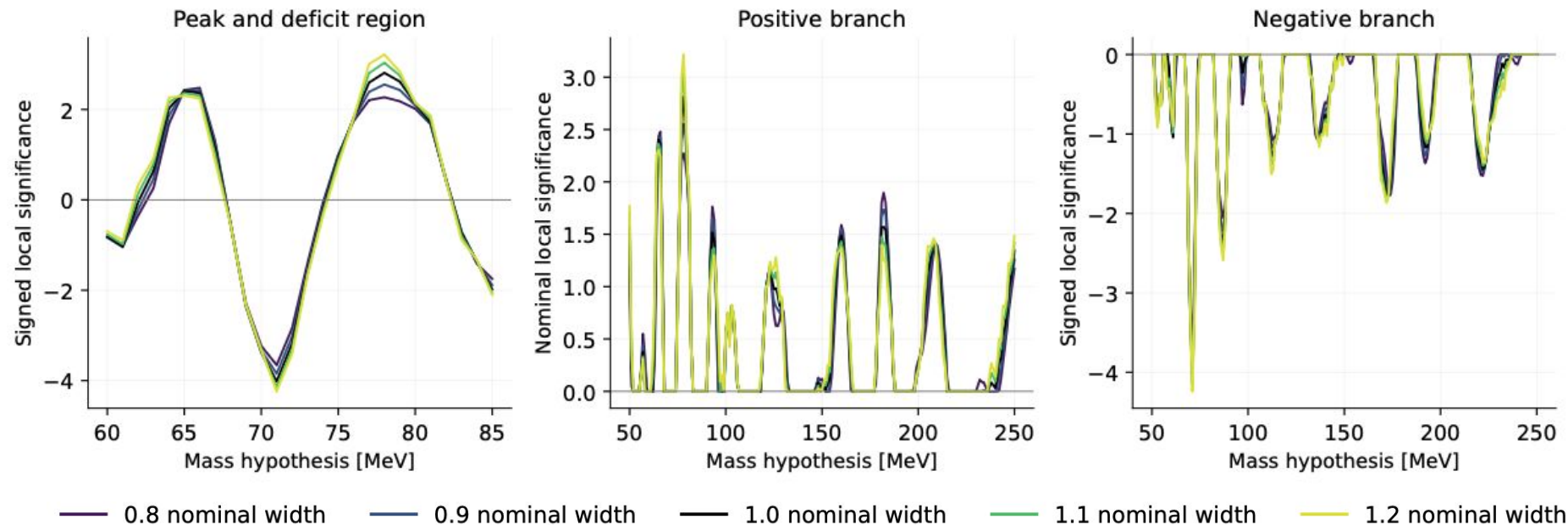
Wiggling the mass resolution pt 1

Impacts

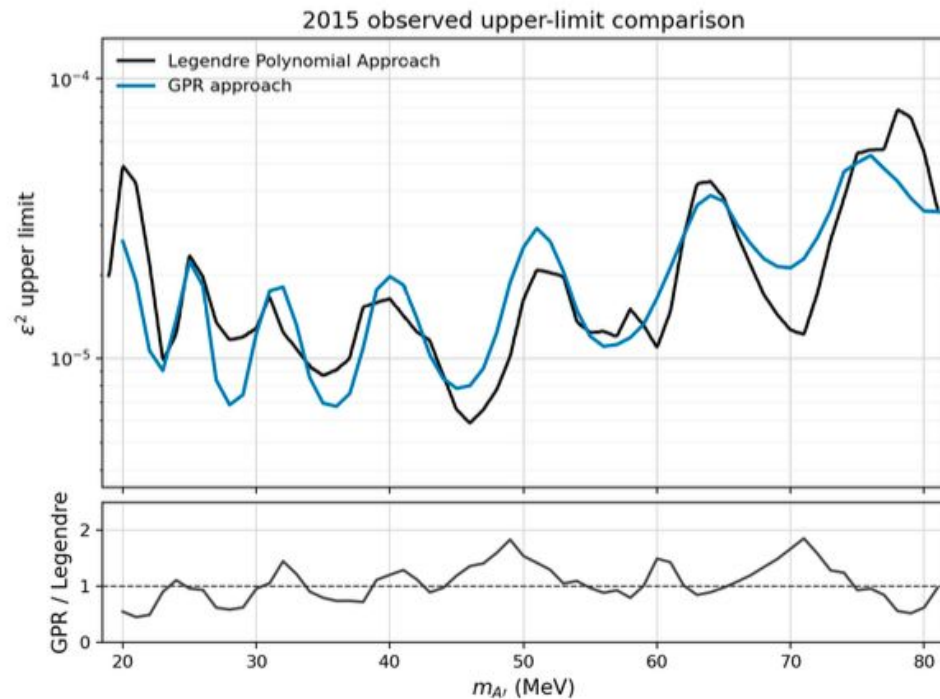
- blind window size
 - (fit uncertainty)
- signal width
- bkg density calculation (marginally)



Wiggling the mass resolution pt 2



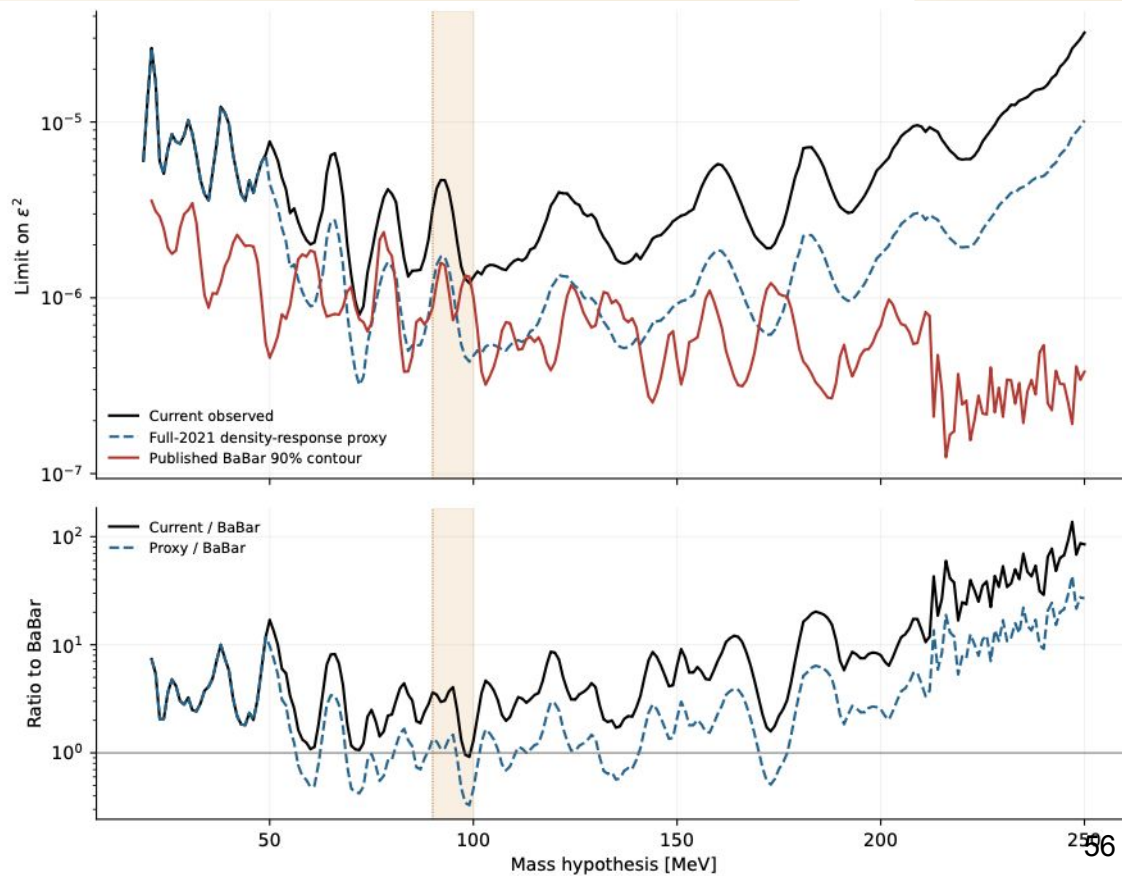
2015 Comparison



Babar Comparison

Currently have small
region of new exclusion

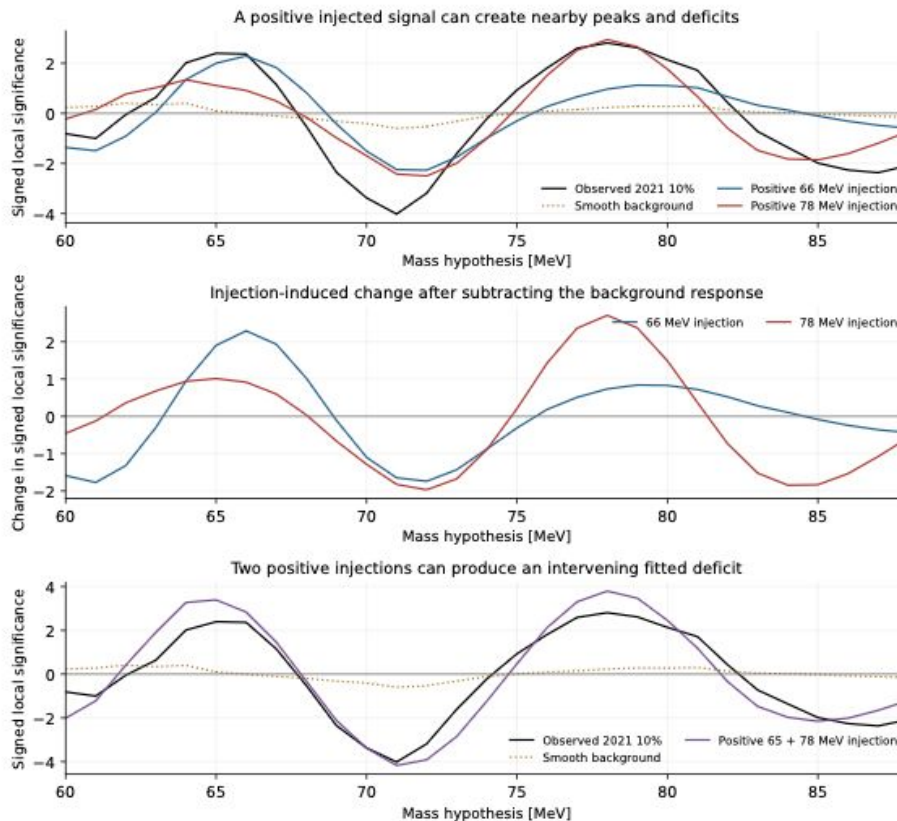
- Exciting!



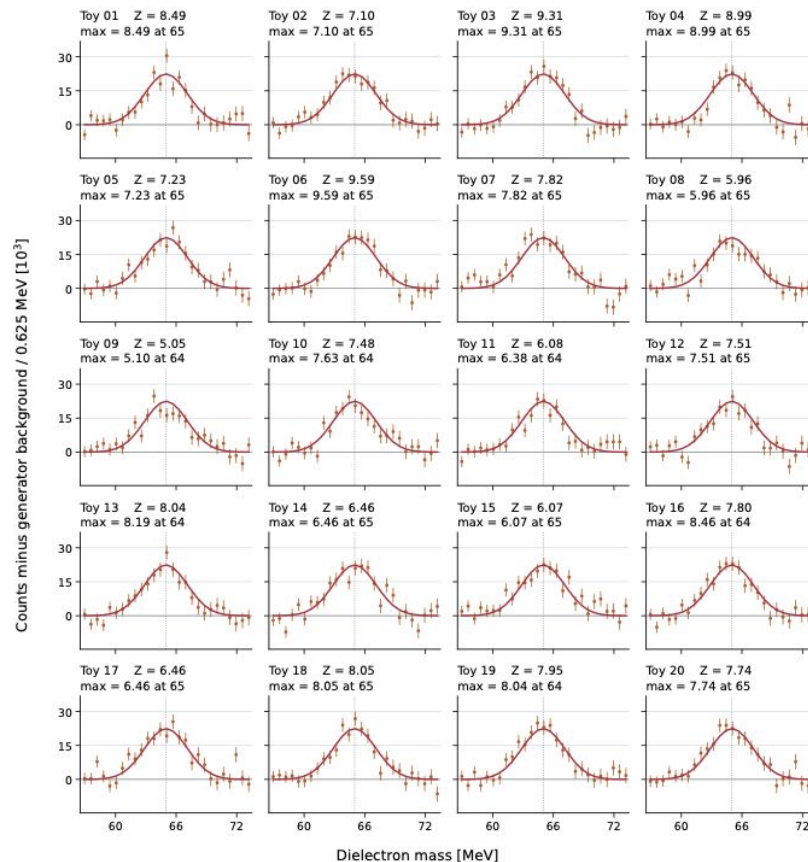
Two Signal Modeling

Two signals can create constructive/destructive effects in observed upper limit

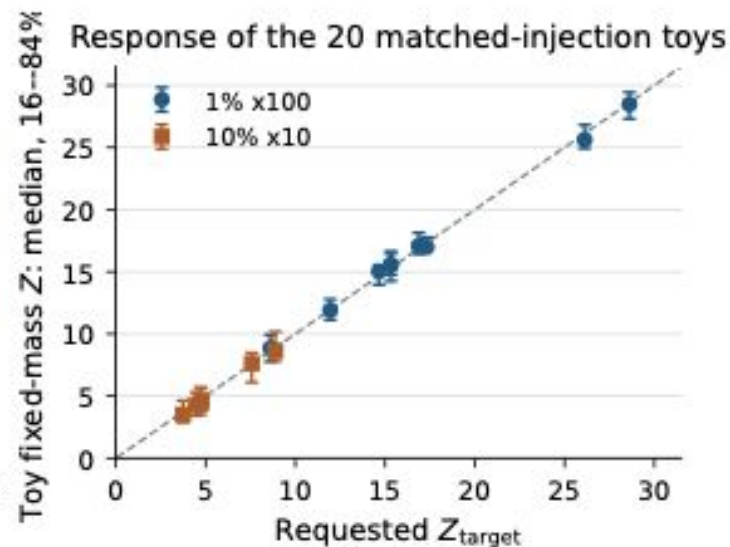
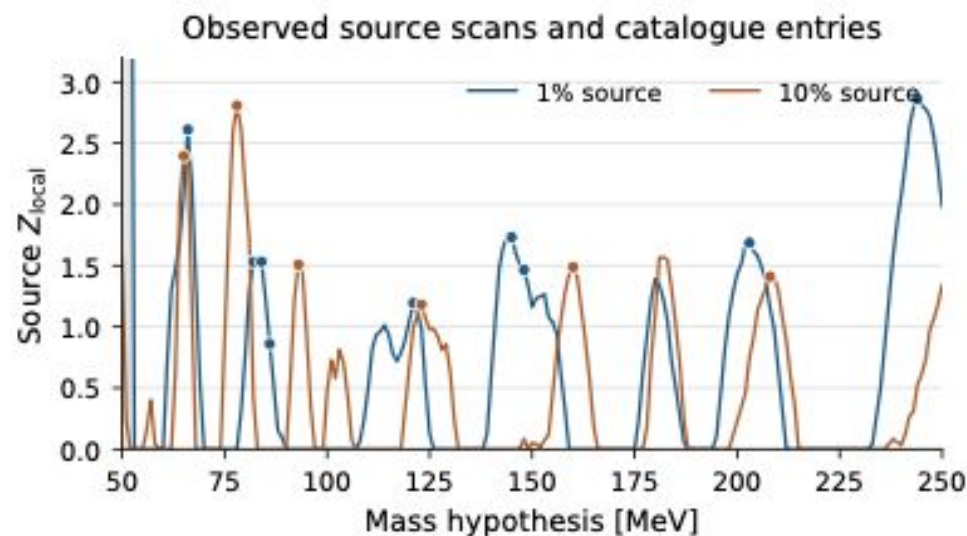
- 2 mild excesses could create deeper deficit than a single nearby signal



hypothetical fit catalogue (65 MeV toys)



Catalogue overview



All “Candidates”

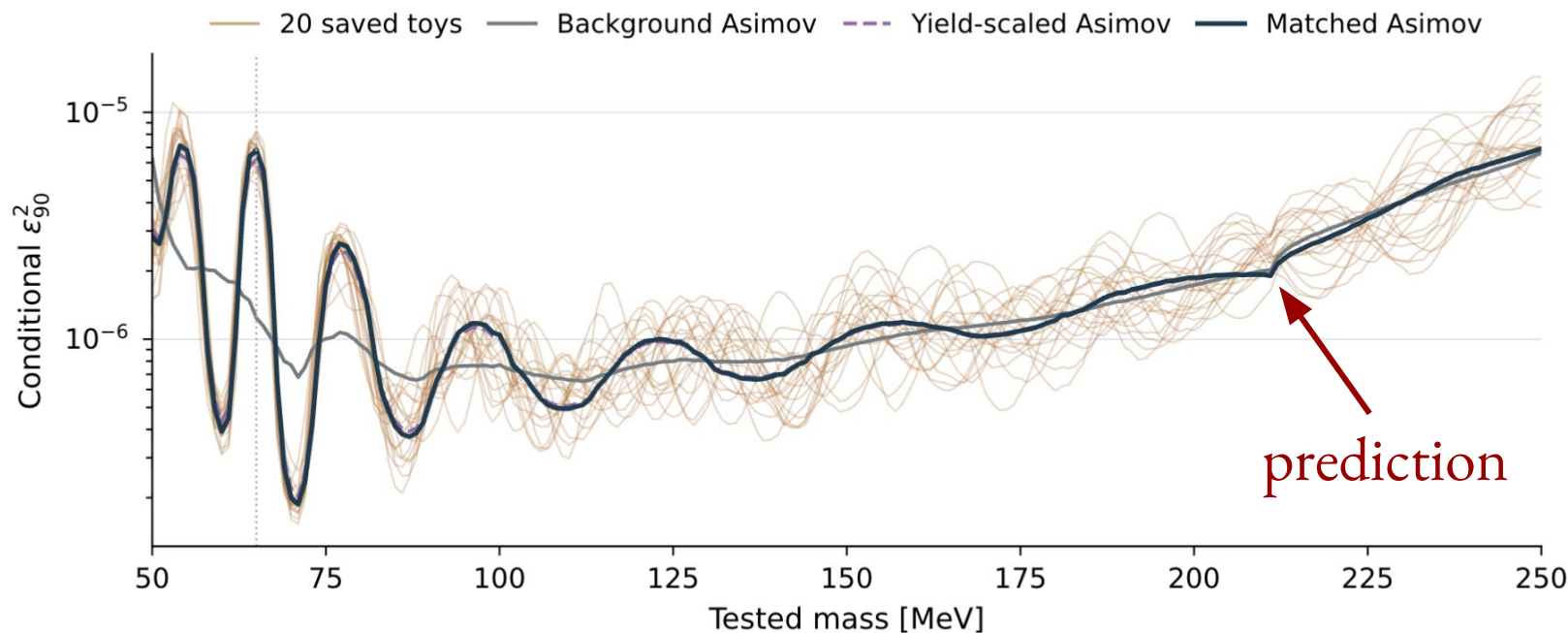
Source	Region [MeV]	m_0 [MeV]	Z_f	Target	Z_A^{yield}	R_a	Toy median [16,84%] fixed mass	Max ₂₀ fixed mass
1%	65	66	2.61	26.12	23.43	1.12	25.62 [24.93, 26.82]	27.79
10%	65	65	2.40	7.58	6.76	1.12	7.63 [6.10, 8.47]	9.59
1%	75	82	1.53	15.32	13.99	1.10	15.54 [14.25, 16.47]	17.92
10%	75	78	2.81	8.88	8.39	1.06	8.64 [7.80, 10.18]	11.62
1%	92	86 [†]	0.86	8.61	7.99	1.08	8.86 [7.77, 9.92]	10.45
10%	92	93	1.51	4.77	4.45	1.07	4.63 [3.92, 5.58]	6.33
1%	120	121	1.19	11.94	11.13	1.07	11.91 [11.13, 12.79]	13.04
10%	120	123	1.18	3.74	3.64	1.03	3.51 [2.89, 4.63]	4.92
1%	160	148 [†]	1.47	14.66	13.59	1.08	15.03 [13.96, 15.53]	16.53
10%	160	160	1.49	4.71	4.54	1.04	4.39 [3.50, 5.71]	7.35
1%	210	203	1.69	16.87	15.50	1.09	17.09 [16.43, 18.13]	19.02
10%	210	208	1.41	4.46	4.16	1.07	4.23 [3.41, 5.30]	6.90
1%	+84	84	1.53	15.33	14.05	1.09	15.57 [14.75, 16.65]	17.05
1%	+145	145	1.73	17.32	16.07	1.08	16.98 [16.78, 17.76]	18.77
1%	+244	244	2.86	28.61	25.31	1.13	28.48 [27.31, 29.45]	30.30

Predictions of observed upper limit

following catalogue shows that we are able to predict shapes of upper limits to first order!

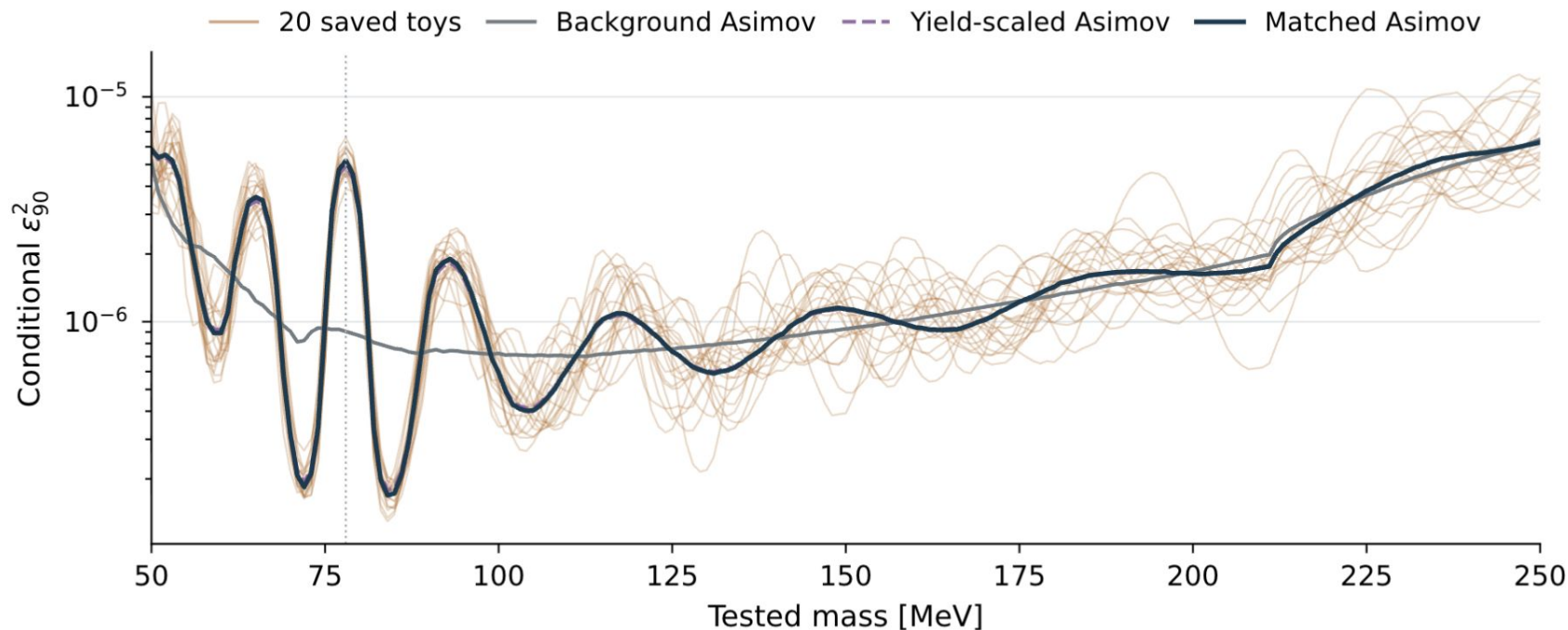
100% Prediction: 65 MeV

Median Projected Local Significance: 7.6 Sigma



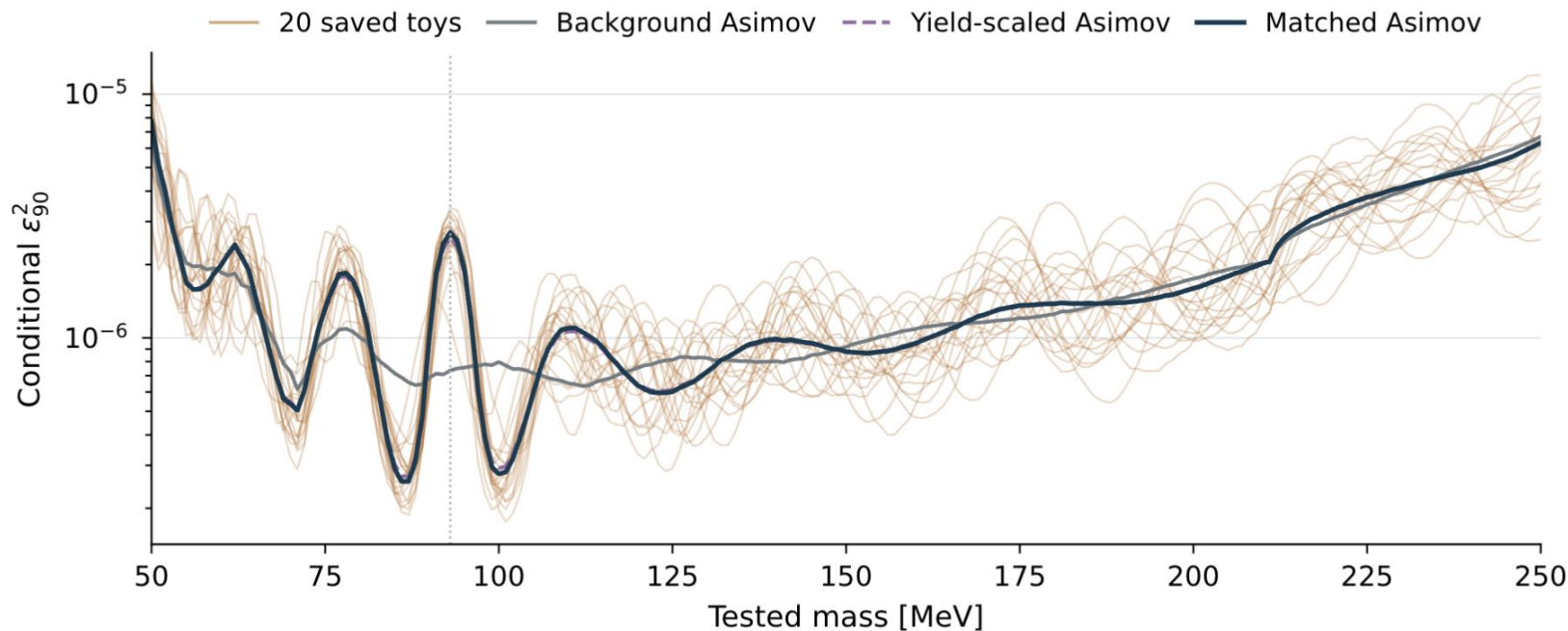
100% Prediction: 78 MeV

Median Projected Local Significance: 8.88 Sigma



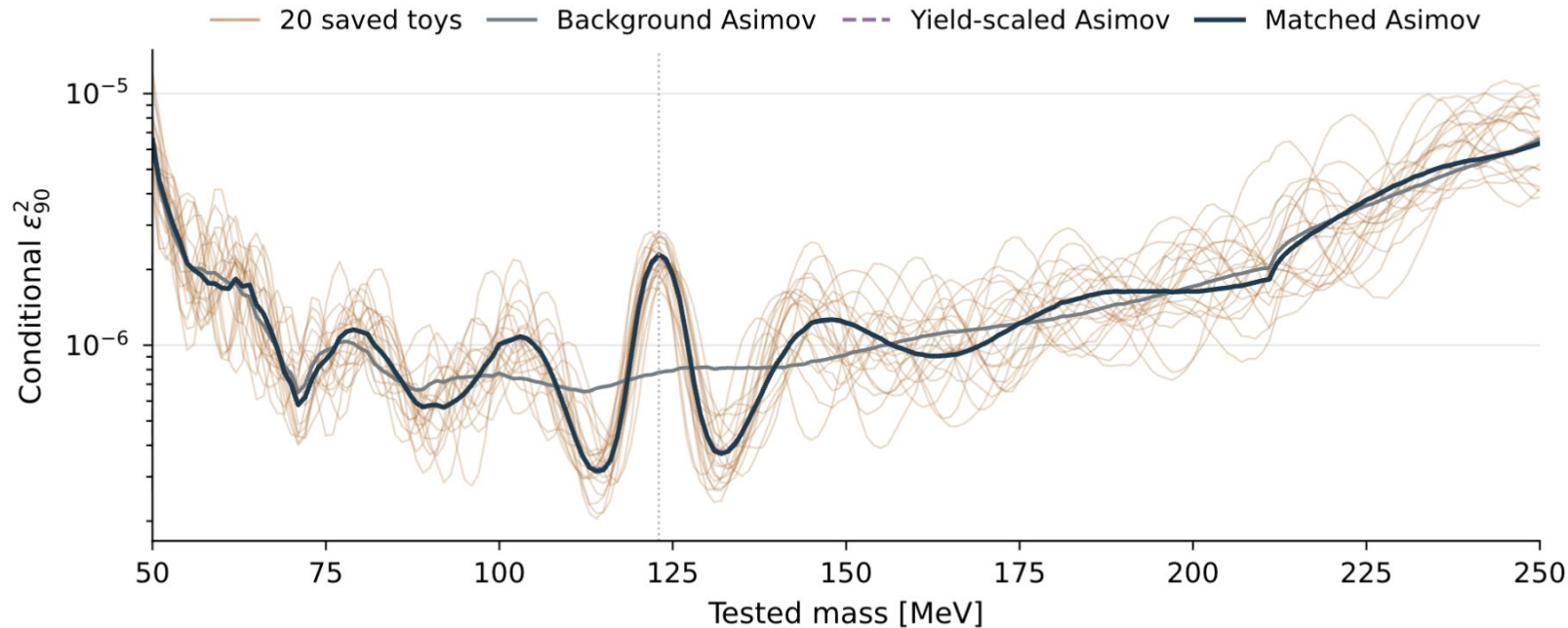
100% Prediction: 93 MeV

Median Projected Local Significance: 4.77 Sigma



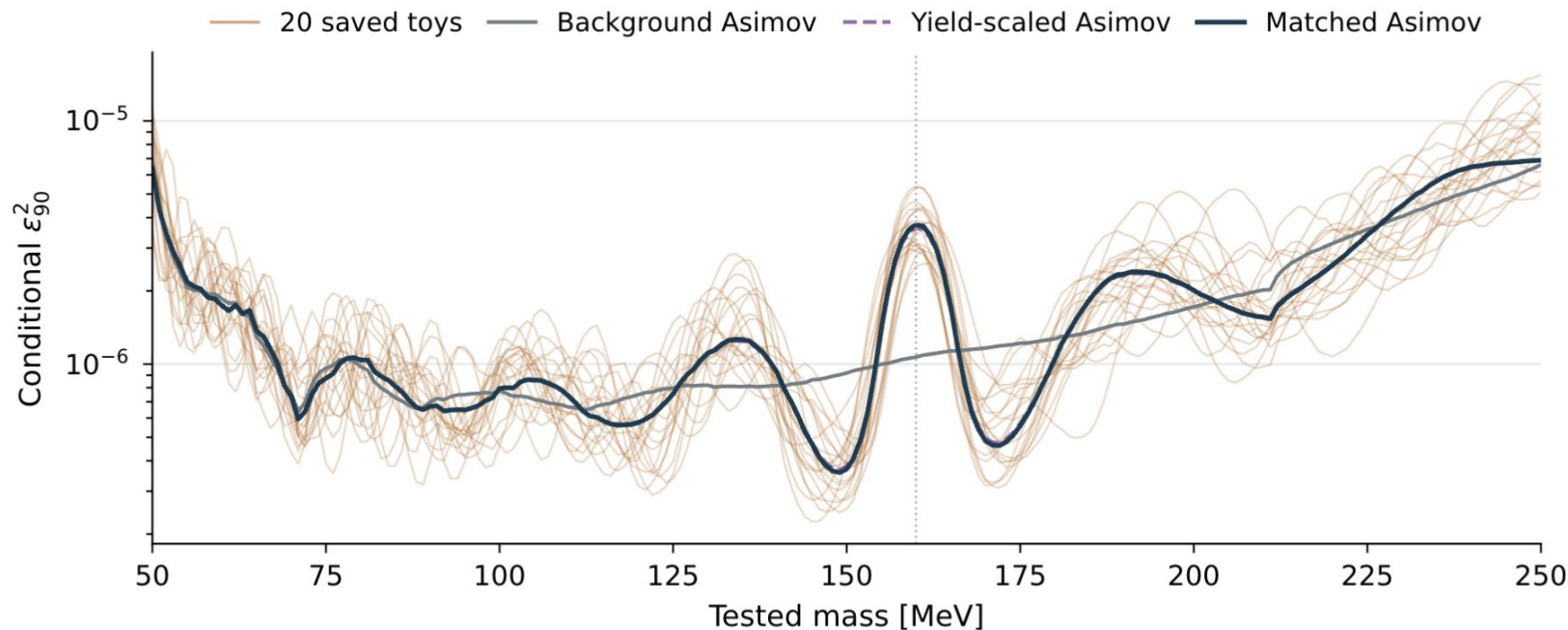
100% Prediction: 120 MeV

Median Projected Local Significance: 3.74 Sigma



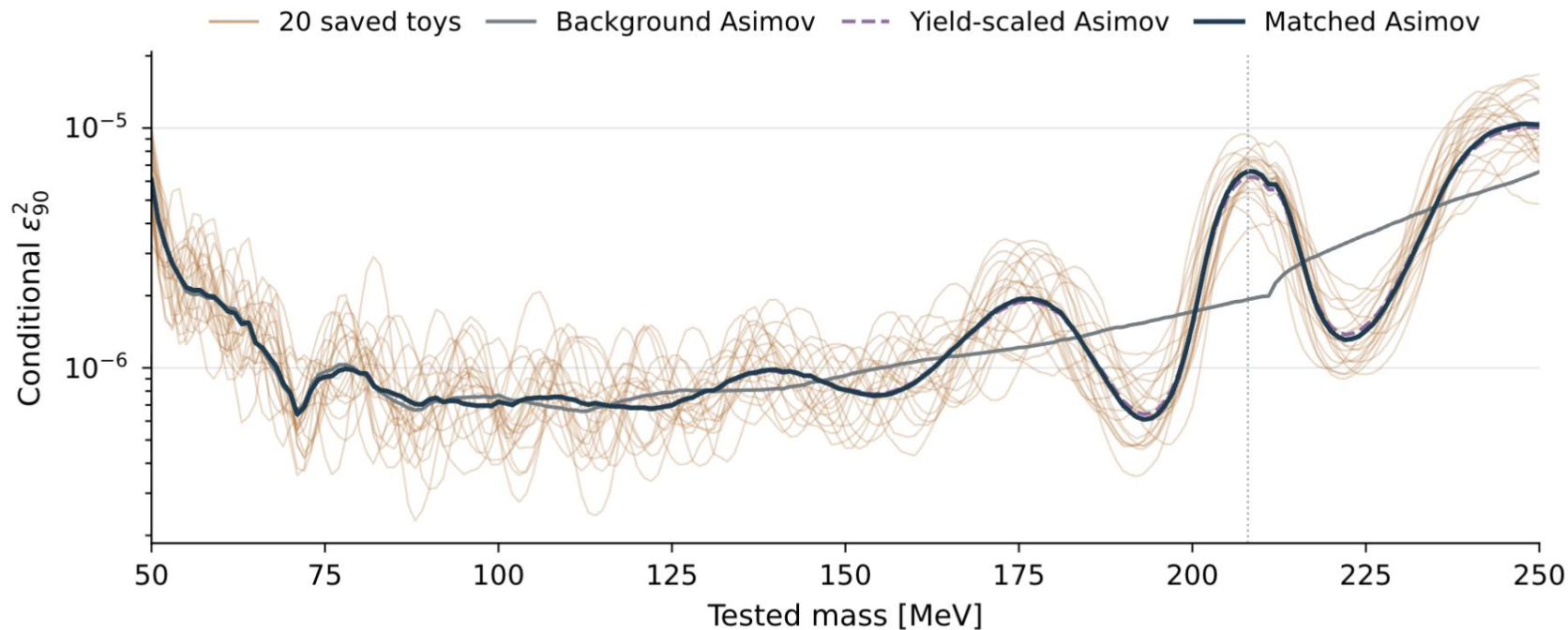
100% Prediction: 160 MeV

Median Projected Local Significance: 4.71 Sigma



100% Prediction: 208 MeV

Median Projected Local Significance: 4.46 Sigma



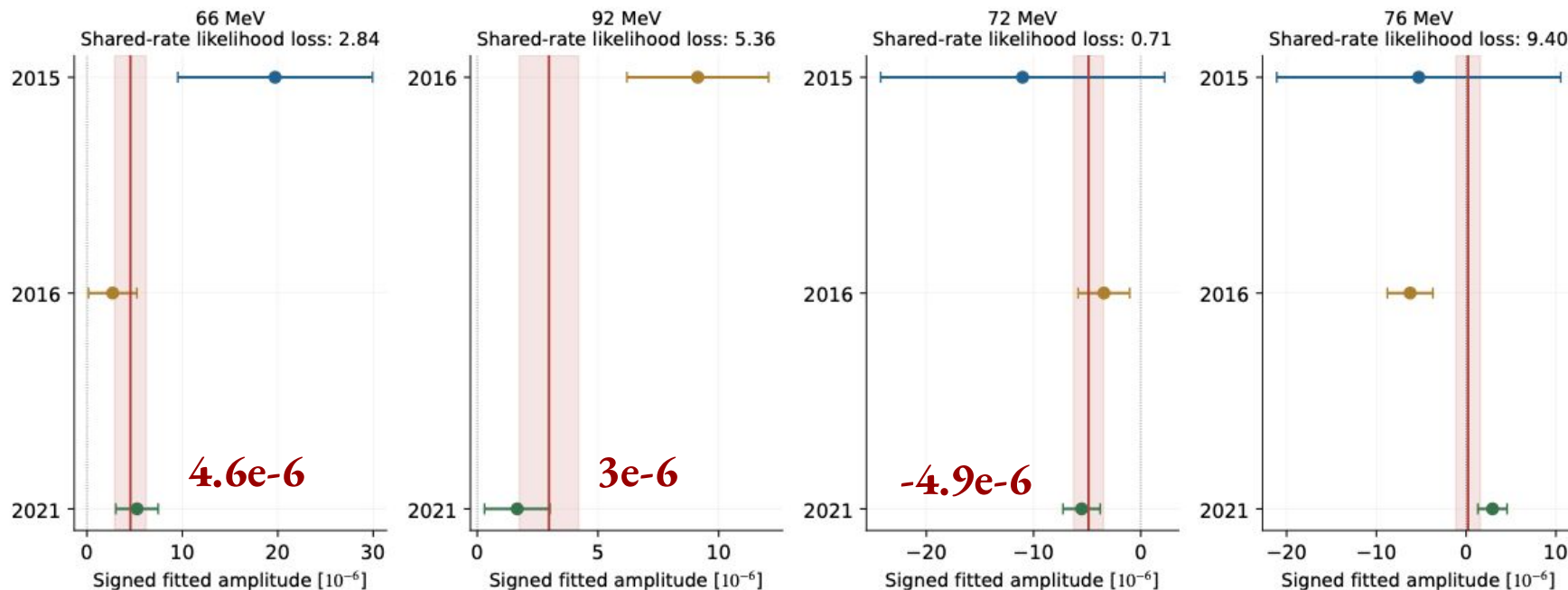
Interpretations

Loss metric

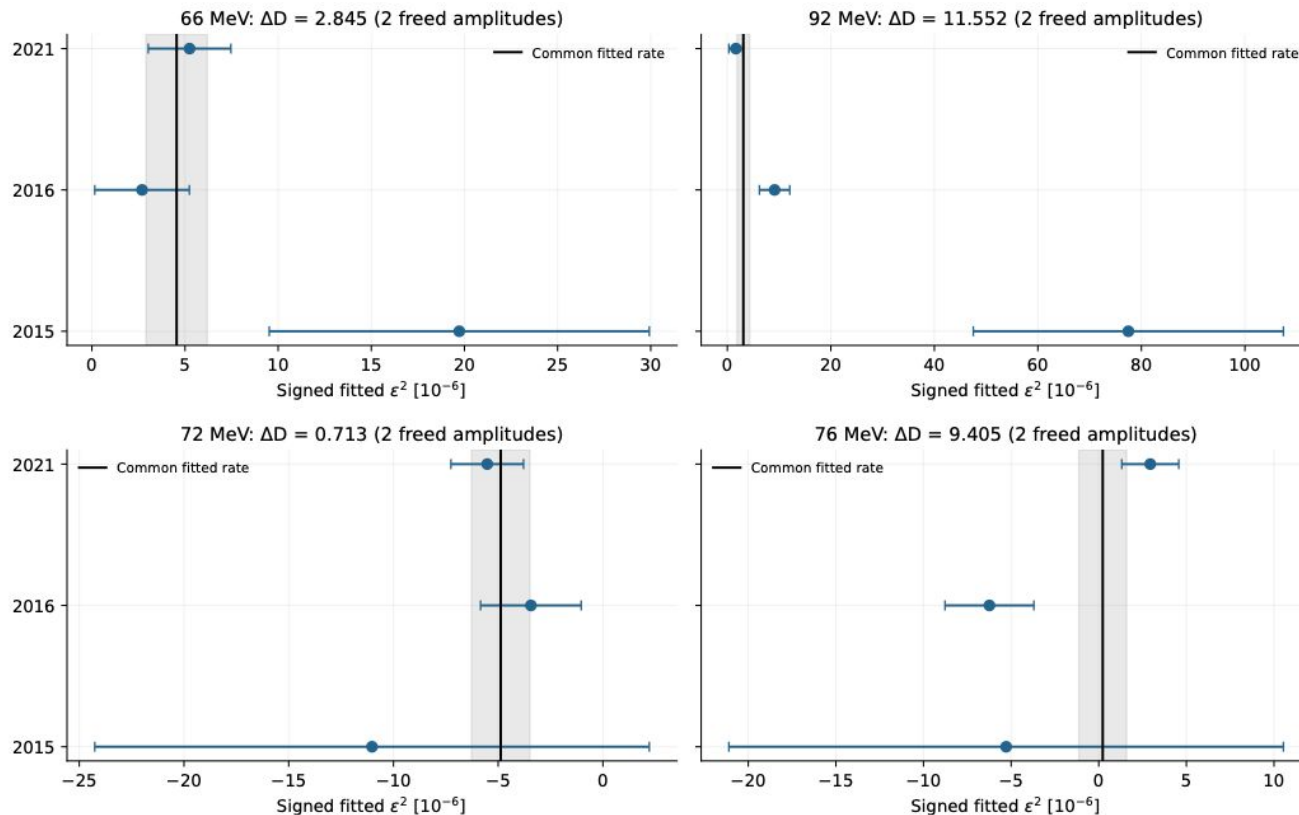
- large loss implies individual datasets describe observation better than shared coupling

$$\Delta D = 2 \left[\text{NLL}_{\text{common}} - \sum_d \text{NLL}_{d,\text{independent}} \right]$$

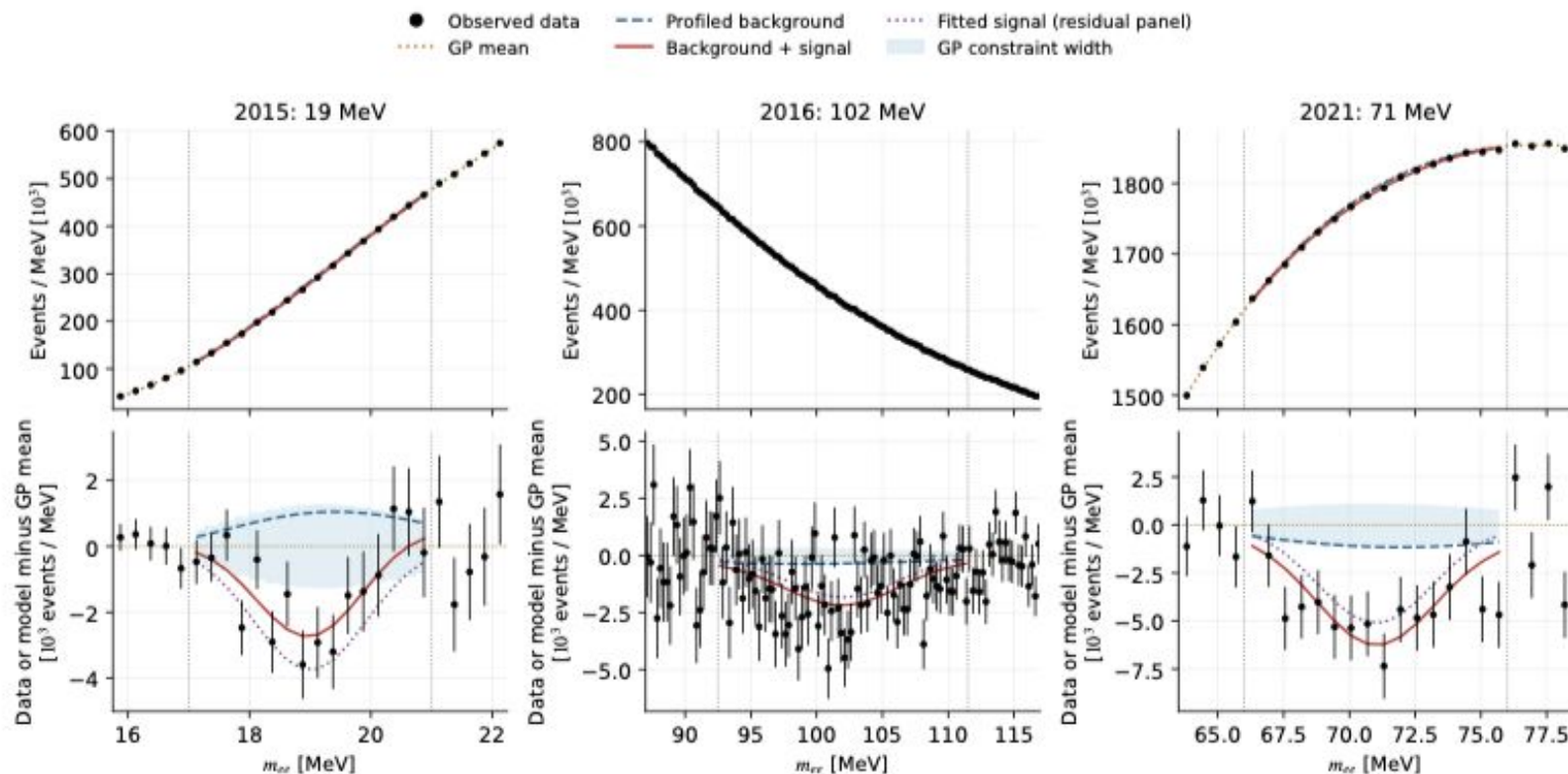
Common rate before 2015 Extension



Common rate after 2015 extension



Leading Deficits



Signal Model for alternative frameworks

Interpolated bkg

Beam Energy

$$\lambda_i = b_i + L_i \theta_i + \mu K_i(m_i, \sigma_i) (E_i / 2.30 \text{ GeV})^\beta$$

power law

Radiative Fraction + dataset dependent info

Profiled Background

The diagram illustrates the components of the signal model equation. Red lines connect the following labels to their corresponding parts in the equation: 'Interpolated bkg' points to b_i ; 'Beam Energy' points to E_i ; 'power law' points to the exponent β ; 'Radiative Fraction + dataset dependent info' points to the function $K_i(m_i, \sigma_i)$; and 'Profiled Background' points to the term $L_i \theta_i$.

Things to consider post-unblinding

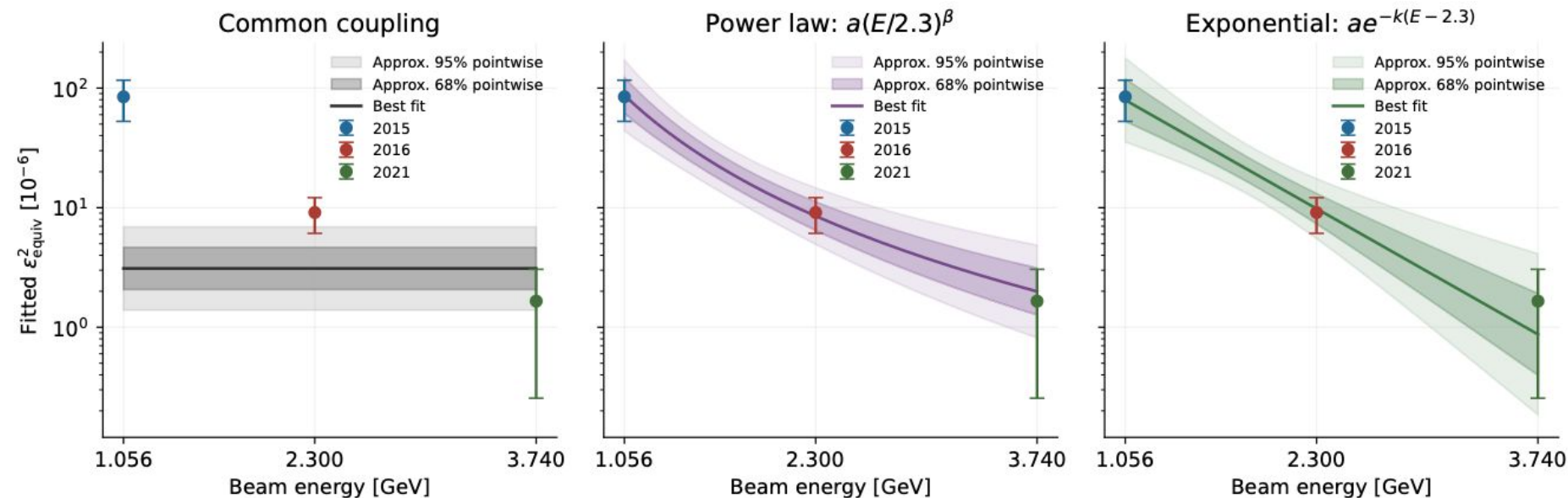
For a fixed common region of interest at 92 MeV

$$Q_G = \sum_i \max(z_i, 0)^2$$

Model or rule	$\sqrt{Q_G}$	Local p_G	Z_G
Common coupling	2.447	7.21×10^{-3}	2.447
Independent positive amplitudes	4.195	1.33×10^{-4}	3.646
Fitted power law	4.182	6.90×10^{-5}	3.812
Fitted exponential	4.148	7.77×10^{-5}	3.782
Equal-weight signed Stouffer	–	3.71×10^{-5}	3.962
Fisher of signed one-sided probabilities	–	6.91×10^{-5}	3.811

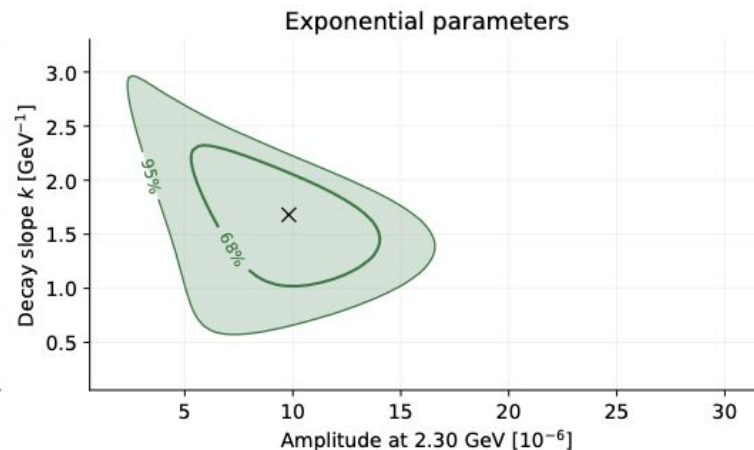
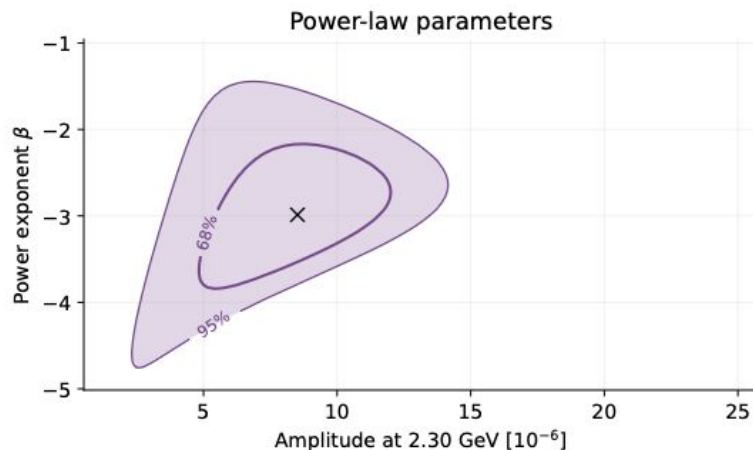
Alternative Combination Methods

Beam energy and Rate normalization assumptions

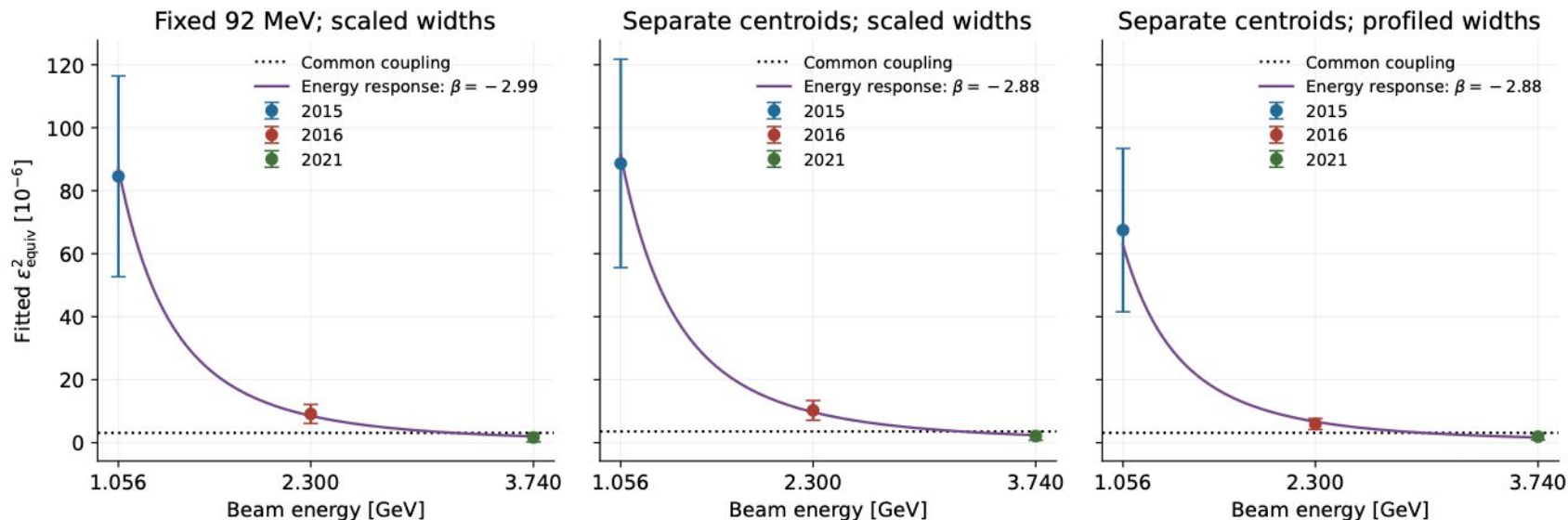


Two Parameterizations

$R(E)$	$\hat{a}$	conditional slope interval, 68% profile
1	3.102×10^{-6}	none
$(E/2.30)^\beta$	8.516×10^{-6}	$\beta = -2.987 [-3.517, -2.470]$
$\exp[-k(E - 2.30)]$	9.817×10^{-6}	$k = 1.681 [1.257, 2.087] \text{ GeV}^{-1}$



Profiling centroid and widths

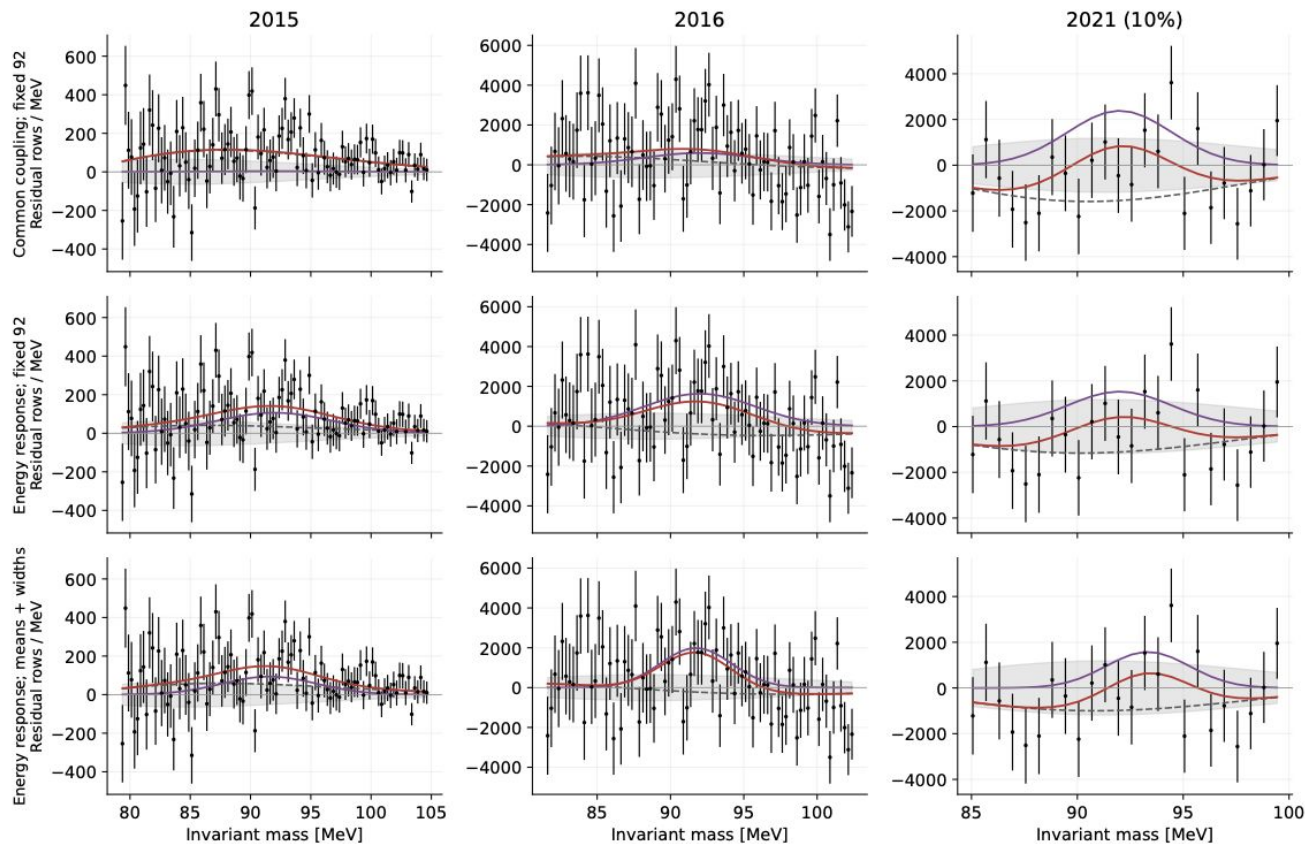


The Different Scenarios

Shared Coupling

Rate Floating
within Energy
Normalization

Rate Floating
within Energy
Normalization+
Mass Resolution
Profiled



Updates and Studies Since Initial Unblinding (08/04/2026)



- I. Geometric Acceptance Study for 2015/2016/2021
- II. Pseudoexperiments of various Statistics
 - A. 1% fit data scaled to 10% and 100%
 - B. 10% fit data fit as 10% and scaled to 100%
- III. 2021 GP fit support optimization
 - A. rising edge
- IV. New observed limit results and all combinations of limits
- V. 2016 100% Comparison
- VI. Contour revitalization (15/16/21 and 15/16/19/21 Projections)
- VII. GPR Echo characterization across all three datasets
 - A. individual/combined dataset search correlations
 - B. predictions for 100% unblinding
- VIII. Scaling Mass Resolution Study
- IX. Profiling Mass resolution as a nuisance
- X. Beam Energy relationship to extracted regions of interest
- XI. “Two-bump” hypothesis